



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Introduction to AI for SRS

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My preferred taxonomy

(1) Symbolic AI
(logic, ontologies, rule-based)

Learning AI

(2) Shallow Machine Learning

(3) Deep Learning

YOUR BIG DATA + Statistics

**YOUR BIG DATA + Statistics +
Neural networks + Optimization**

Generative AI

**BIG DATA(PROVIDER'S and YOUR) + Statistics
+ Neural networks + Optimization**

(4) ChatBot Drafting content Writing code Generating image, video ...

(5) AI agent

(6) Agentic AI



Models

- GPT-4 / GPT-4o (*OpenAI*): Unknown parameters(estimated >1T, Mixture-of-Experts)
- Focus on multimodality, complex reasoning, and state-of-the-art performance
- Claude (*Anthropic*): Family of models (*Haiku*, *Sonnet*, *Opus*) with different performance and “personalities”
- Gemini (*Google*): Family of models (*Ultra*, *Pro*) natively multimodal



Open weight and open source models (1/2)

- **Llama 2 & Llama 3** (*Meta*): A family of models (7B, 13B, 70B) that has democratized access to powerful LLMs, becoming a standard.
- **Mistral & Mixtral** (*Mistral AI*): A France startup that has produced smaller but efficient and high-performance models, often using the “Mixture of Experts” (MoE) architecture
- **DeepSeek**: A family of Chinese models with a strong specialization in code generation and text understanding
- **Grok** (*xAI*): A free AI assistant developed by xAI and integrated into X (formerly, Twitter) *to maximize truth and objectivity*. Grok is an open-weight model that offers real-time search, image generation, trend analysis, and is well-known for its direct, “anti-woke,” and often blunt, sarcastic, or rebellious behavior. It was designed by Elon Musk to be a “TruthGPT” alternative to other “soft” AI models. It aims to provide users with unfiltered answers and to mirror the spirit and political leanings of Musk himself



Open weight and open source models (2/2)

- **Open-source software applications** (e.g., Apache 2.0) provide full transparency, including source code, model weights, and training data. They allow unrestricted modification and reproduction
- **Open-weight models** (e.g., Llama 3) provide the trained parameters (*weights*) for download and local use but withhold the training data, architecture details, and training code

Open-source offers complete visibility, fostering trust and community auditing.

Open-weight models are considered **black boxes** because only the final result is available, making it harder to verify for biases especially during the training phase



“Attention is all you need” [Google research, 2017]

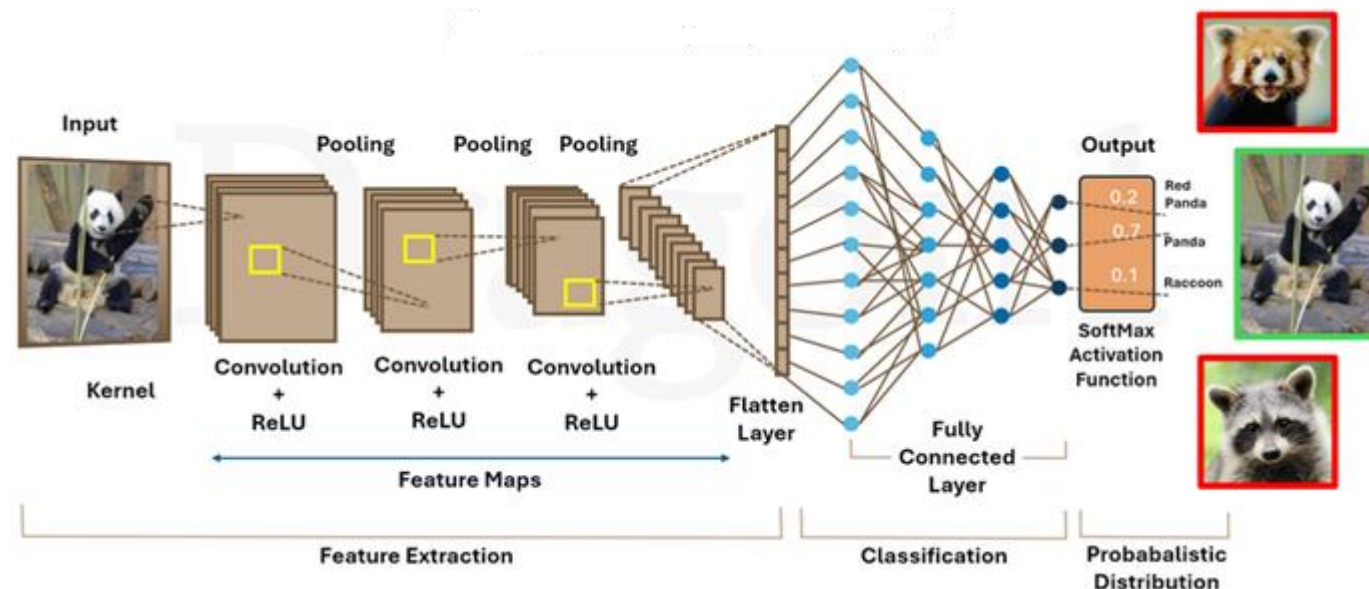
The dominant AI models are based on complex configurations of:

- **Convolutional neural networks** (CNN) in an encoder-decoder configuration of a *fast-forward* neural network
- **Recurrent neural networks** (RNN). RNNs *feed results back* into the network and they are commonly used for natural language processing including machine translation, speech analysis, and sentiment analysis



Convolutional neural networks (CNN)

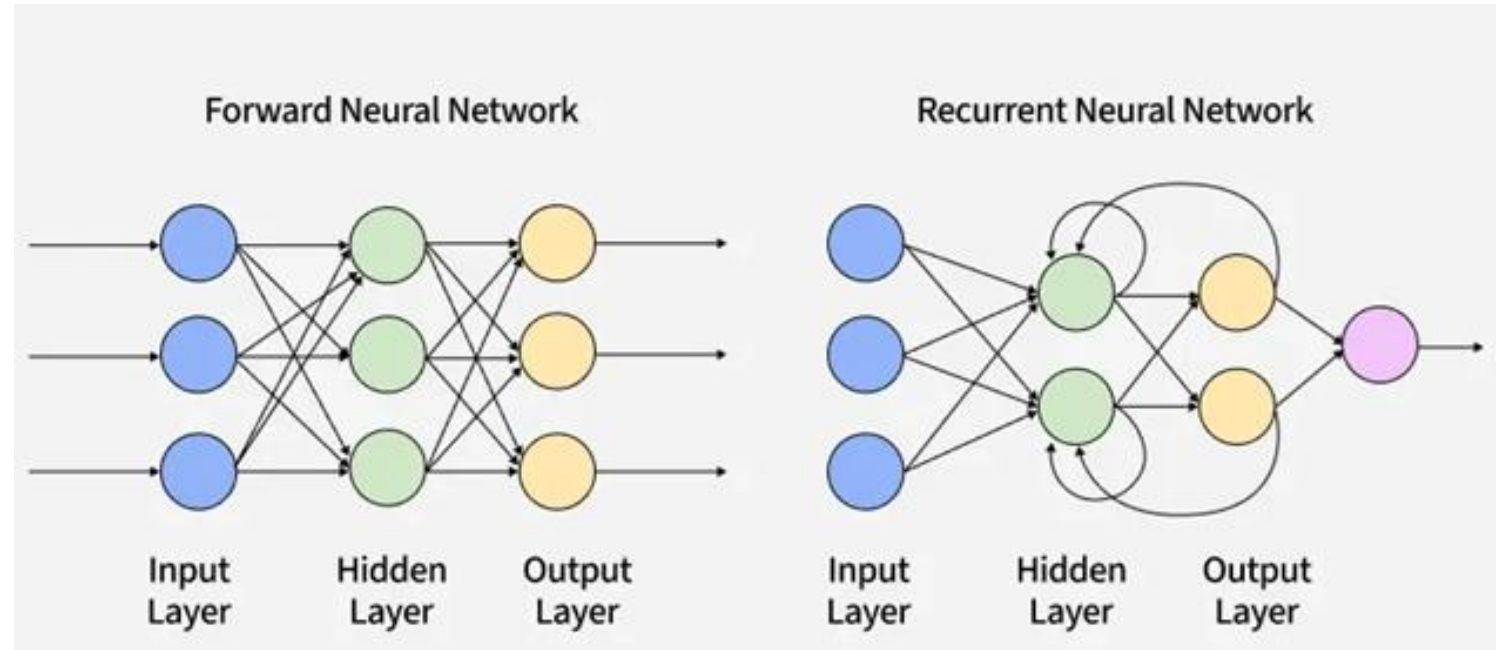
CNNs are *feedforward* neural networks that use filters and pooling layers. They are commonly used to solve problems involving spatial data, such as *image classification* including facial recognition and medical analysis



Recurrent neural networks (RNN)

RNNs *feed results back* into the neural network

- They are better suited to analyzing temporal and sequential data, such as text or videos. For this reason, they are commonly used for **natural language processing** including machine translation, speech analysis, and sentiment analysis

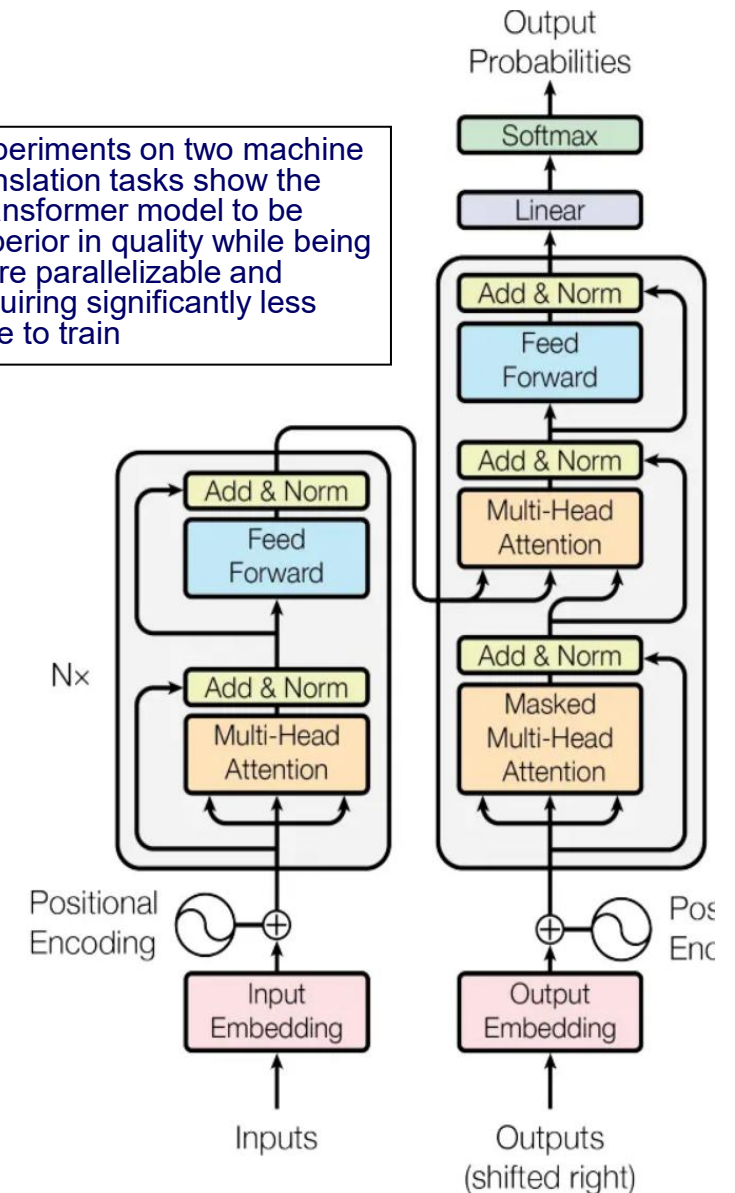


“Attention is all you need” [Google research, 2017]

Google researchers propose a new network architecture, the **Transformer, based solely on attention mechanisms**, removing the complexity of creating combinations of RNN and CNN architectures. The Transformer model is built around a few key components:

- **Encoder-Decoder Structure**
- **Encoder** processes the input sequence
- **Decoder** produces the output sequence
- Each part contains multiple **layers**, and both the encoder and decoder use a mechanism called **self-attention** to process their respective inputs
- **Self-attention is the core idea of this model.** Instead of processing data sequentially (like RNNs), the model looks at all the words in a sentence simultaneously. It then determines which words are most relevant to each other at every layer of the network. This allows the model to focus more on important parts of the input and capture relationships between distant words

Experiments on two machine translation tasks show the Transformer model to be superior in quality while being more parallelizable and requiring significantly less time to train



The fundamental questions

Where *can* AI provide value, where it *does not* (yet),
and where *we do not know* if it will ever provide value?



Where does AI work very well? (*super-human results*)

Games: Chess (1996), Jeopardy (2011), Go (2015/2016),
Poker (2016/2017), Diplomacy (2022), ...

e-games



We didn't tell how to play poker. We gave him the rules of poker and said "learn on your own" (He learned to cheat - incomplete information)

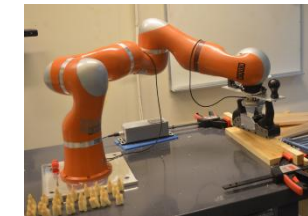
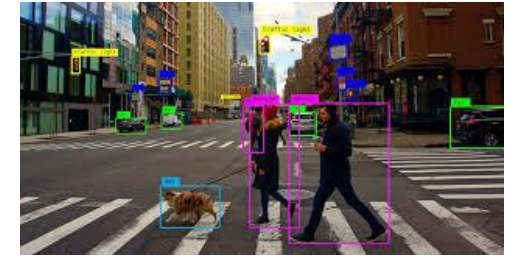
A seven-player game much harder than a two-player game. You must consider many possible strategies of not one but six other players

***The model and the real world coincide
Moreover, rules and constraints exist***

Where does AI work well?

(results comparable to humans or better)

- Computer vision
- Facial recognition
- Natural Language Processing, Text recognition, Automatic translation, Chatbot
- Speech recognition
- Medical diagnostic imaging
- Profiling, Recommendation system
- Autonomous robots (including vehicles) in controlled contexts



The real world is different but not so different
Moreover, they have a lot of data for training

Common characteristics of the working applications

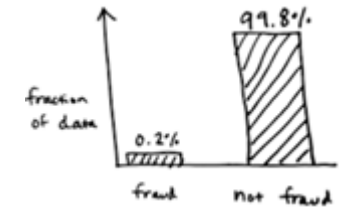
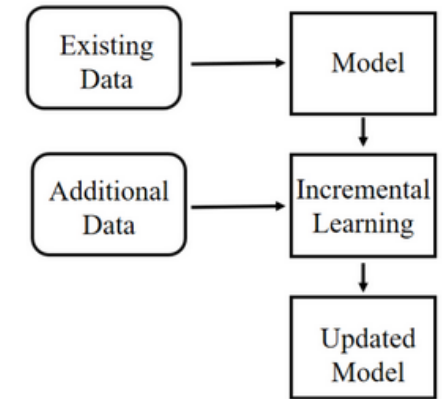
- Games: Chess, Go, Poker, ...
- e-games
- Computer vision
- Facial recognition
- Natural Language Processing, Text recognition, Automatic translation, Chatbot
- Speech recognition
- Medical diagnostic imaging
- Autonomous robots (including vehicles) in controlled contexts

- 1) Lots of good and relatively stable data**
- 2) Clear goal(s)**
- 3) Perimeter where the AI model is applied**
- 4) Rule-based**
- 5) No opponent**



Still (*research*) problems in other scenarios

- Continuously varying scenarios (they would require ***continuous learning*** – *ongoing research*)
- Contexts that cannot be easily modeled or delimited
- Contexts where rules are not defined or do not exist
- Independent relationships between input and output variables
- Training samples are **unevenly distributed**: some events are the large majority, and others are rare (e.g., in security and cybersecurity, malicious samples are much scarcer than the benign samples)
- Contexts in which even a minimal false positive is not acceptable (e.g., *is a 99.98% accuracy acceptable?*)
- Autonomous robots (including vehicles, AGV, smart objects) used in the real world



Example: Self-driving and Driving

Self-driving systems are built based on training over millions of kilometers driven → Faced with a new situation, they may have problems (*hallucinations*)



Humans get a license after few driving lessons, and can drive everywhere → **We can use intuition to address unseen issues and situations.** Moreover, we know how to manage risks and taking **liability**

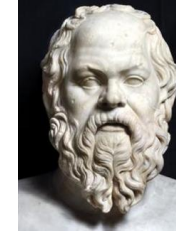


Four stages of human competence

1) **Unconscious incompetence** (*Dunning-Kruger effect*: “ignorance can generate more self-confidence than knowledge”)



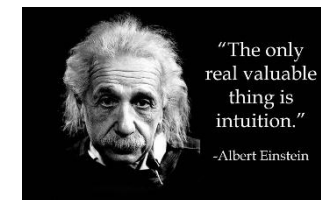
2) **Conscious incompetence** (*Socrates*: “the true wise man is he who knows that he does not know, because only he who knows his own ignorance is interested in knowing”)



3) **Conscious competence** (the professionals that apply what they have learned to the *known* cases)

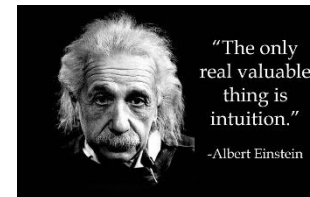
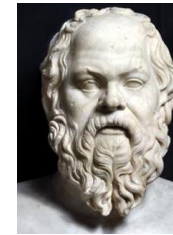


4) **Unconscious competence** (the professionals that are able to apply what they have learned even to *unknown* cases)



Humans

- 1) **Unconscious incompetence** (*Dunning-Kruger effect*: “ignorance can generate more self-confidence than knowledge”)
- 2) **Conscious incompetence** (*Socrates*: “the true wise man is he who knows that he does not know, because only he who knows his own ignorance is interested in knowing”)
- 3) **Conscious competence** (the professionals that apply what they have learned to the *known* cases)
- 4) **Unconscious competence** (the professionals that are able to apply what they have learned even to *unknown* cases)



Machines (no awareness, but ...)



Explicit knowledge



*Tacit knowledge,
Intuition, Instinct*



AI Gone Wrong:

An updated list of AI errors, mistakes and failures

There have been a **myriad of AI success stories**, but there have also been a lot of cases where harnessing AI has gone **horribly wrong**.

Recognizing the limitations and risks surrounding AI tools is an important part of this course

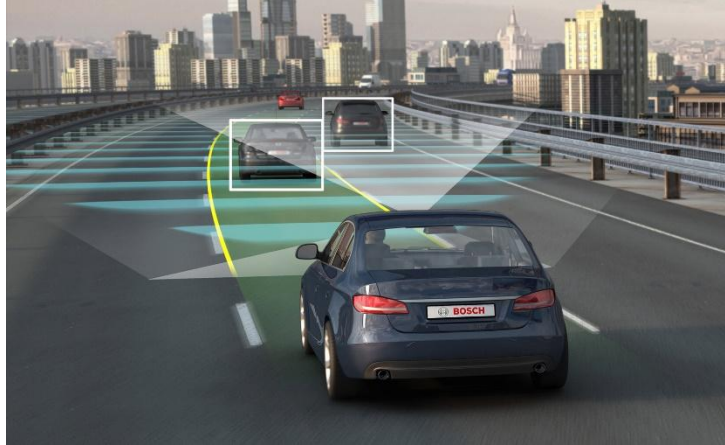
<https://tech.co/news/list-ai-failures-mistakes-errors>



Autonomous systems in a *modified* reality



Autonomous systems in the reality



Large Language Model (LLM)

- Large language models (LLMs) are a type of AI that uses deep learning to understand, generate, and work with human-like text

Gemini



Claude

deepseek

LLM (Abstraction)

Text



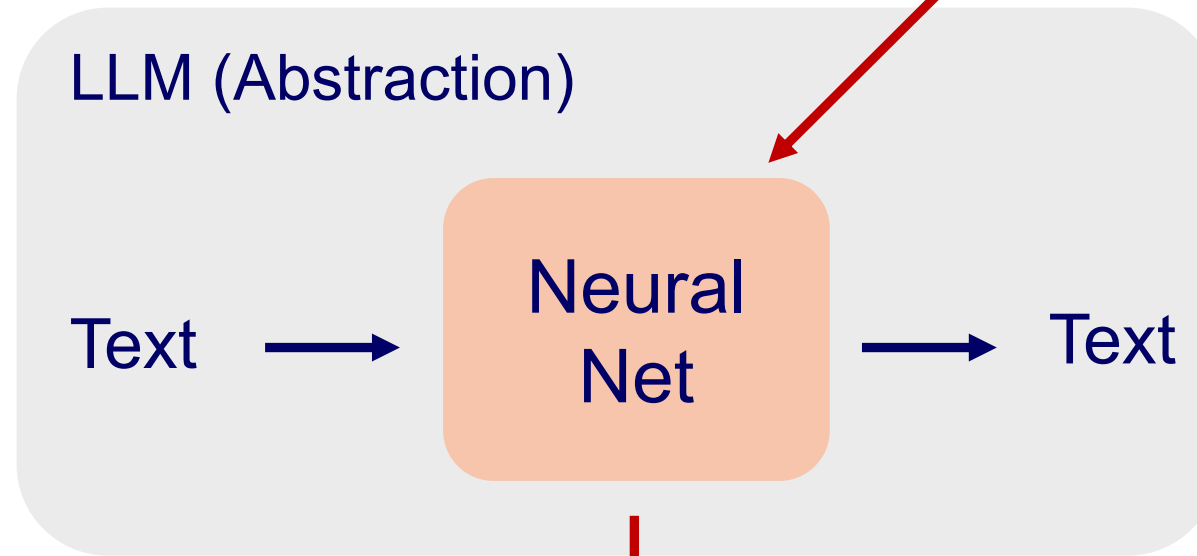
Neural
Net



Text

How to get the Neural Net (*Deep Learning*)?

Very complex. Hard to get such a function!



Train a Neural Net:
Given data, use *gradient descent* to approximate an optimal neural net

Deep Learning is computationally hard

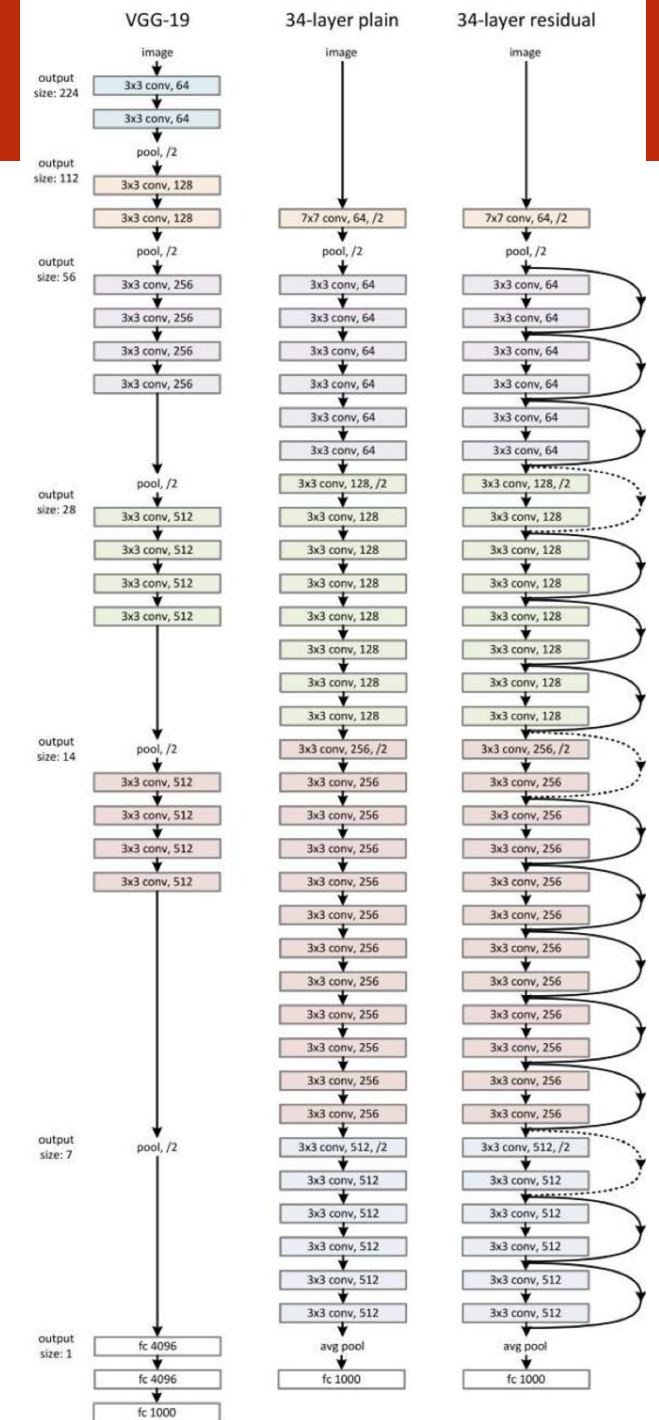
Peta-Exa scale compute ($\sim 10^{18}$)

Size:

- Billions-trillions of operations per sample
- Millions-billions of samples per training
- Deepseek uses 2.79 million GPU hours

Shape:

- Fine-grained: A lot of parallelism;
- presents opportunity to accelerate
- Coarse-grained: Inherently serial



Compute acceleration is always more necessary

1. Faster generation (*Quicker word generation*)
2. Better results (*More compute for reasoning*)
3. Shorter iteration cycles (*Training is faster*)

...

xAI Datacenter at Memphis

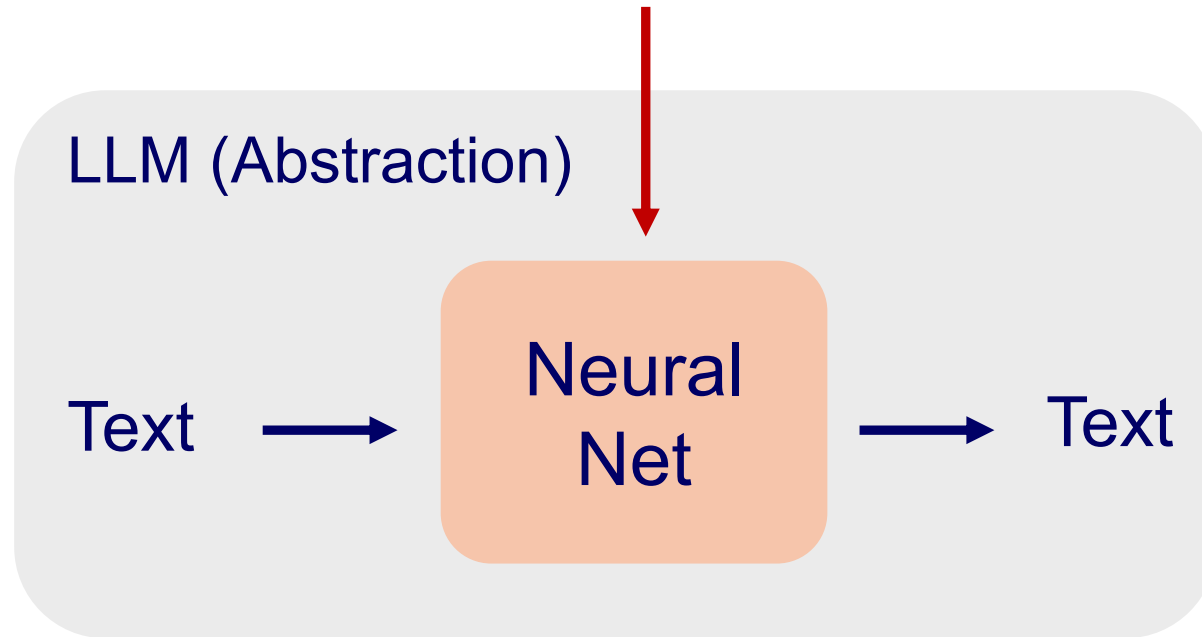


Google TPU cluster

- Proliferating across industries and applications
- Large and growing portion of workload in datacenter
- Between 2012 and 2018 this AI workload grew 300,000x

What model architecture LLM uses?

What does this look like for an LLM?

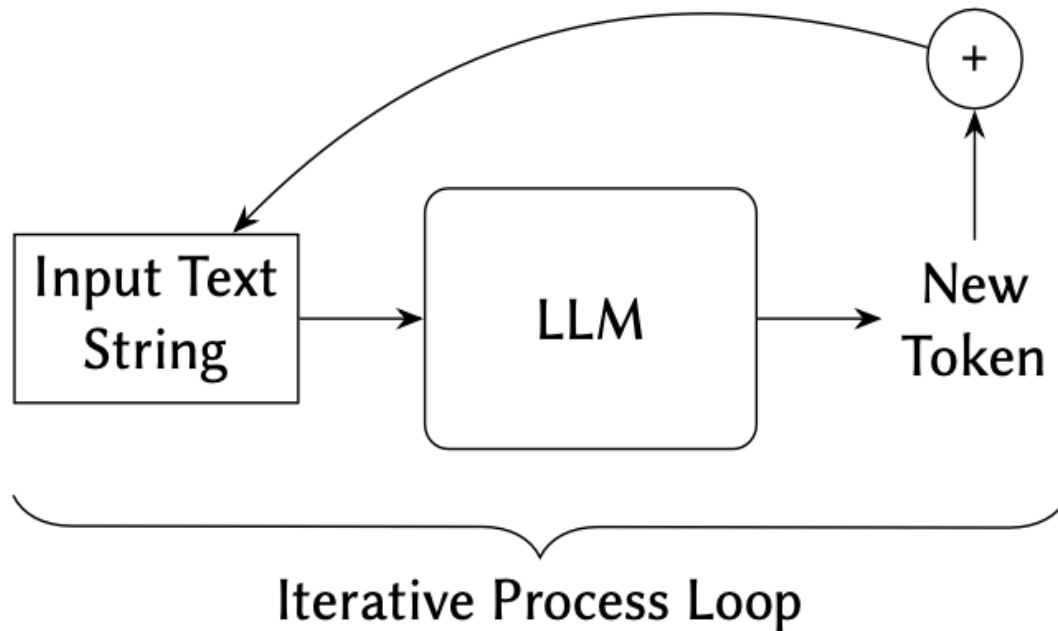


Key ideas:

1. Autoregressive Text Generation
2. Transformer Architecture

Autoregressive text generation process via a transformer

The transformer processes the input text, producing a probability distribution over the potential next tokens using its neural network. A token is stochastically selected from this distribution (in our setting, we force deterministic behavior by setting seeds) and added back into the text, which is then re-fed into the transformer, continuing the iterative text generation process.



1. Predict one token (a word) at a time
2. Add this new token to the input string

Performant but slow



Need compute!



Transformer: A neural net architecture

Basically, a complex function:

$$\text{Transformer}(X , \theta) \rightarrow Y$$

Input X : a sequence of words

Output Y : a sequence of words

Example:

$$\text{Transformer}(\textit{“the robots will bring”} , \theta) \rightarrow \textit{“prosperity”}$$

Tokenization (convert everything to digits):

$$\text{Transformer}([3206, 2736, 3657, 400] , \theta) \rightarrow 2532$$



Transformer: A neural net architecture

The Transformer is a machine learning model for sequence modeling. Given a sequence of *things*, the model can predict what the next *thing* in the sequence might be. In this article, we will look at word sequence prediction, but you can apply Transformers to any kind of sequential data, such as stock price prediction in financial applications, pixels / image frame in visual content generation, or robot motion generation.

Basically, a complex function: $\text{Transformer}(X, \theta) \rightarrow Y$

Input X : a sequence of words - Output Y : a sequence of words

Example: $\text{Transformer}(\text{"the robots will bring"}, \theta) \rightarrow \text{"prosperity"}$

Tokenization (convert everything to digits):

$\text{Transformer}([3206, 2736, 3657, 400], \theta) \rightarrow 2532$



Transformer

The Transformer is a machine learning model for sequence modeling. Given a sequence of *things*, the model can predict what the next *thing* in the sequence might be. In this example, we look at word sequence prediction, but you can apply Transformers to any kind of sequential data, such as stock price prediction in financial applications, pixels / image frame in visual content generation, or robot motion generation

$$\text{Transformer}(X, \theta) \rightarrow Y$$

Basically, a complex function:

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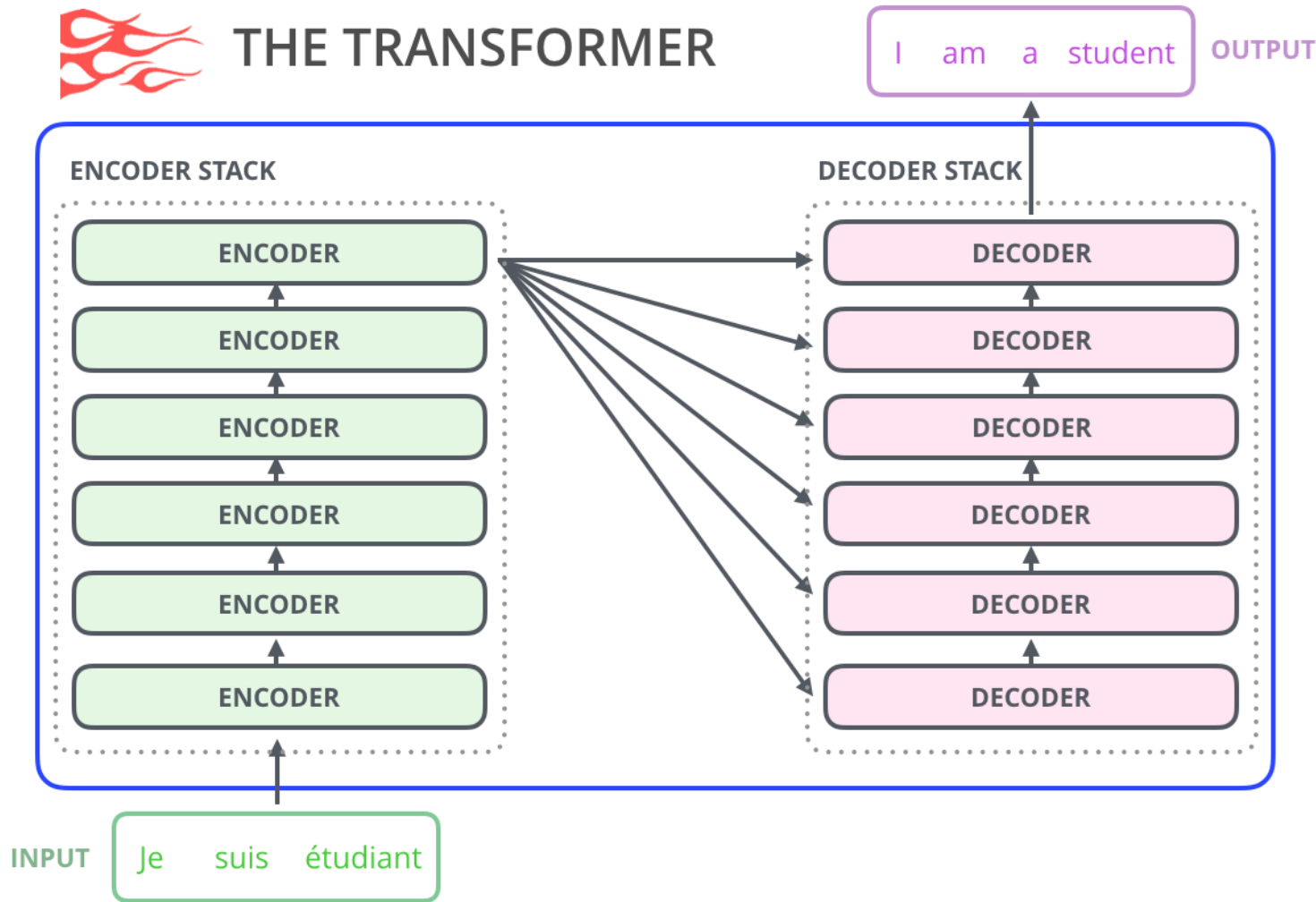
$$\text{Transformer}([3206, 2736, 3657, 400] , \theta) \rightarrow 2532$$



Transformer model architecture

1. Predict one token (a word) at a time

2. Add this new token to the input string



Text is not easy



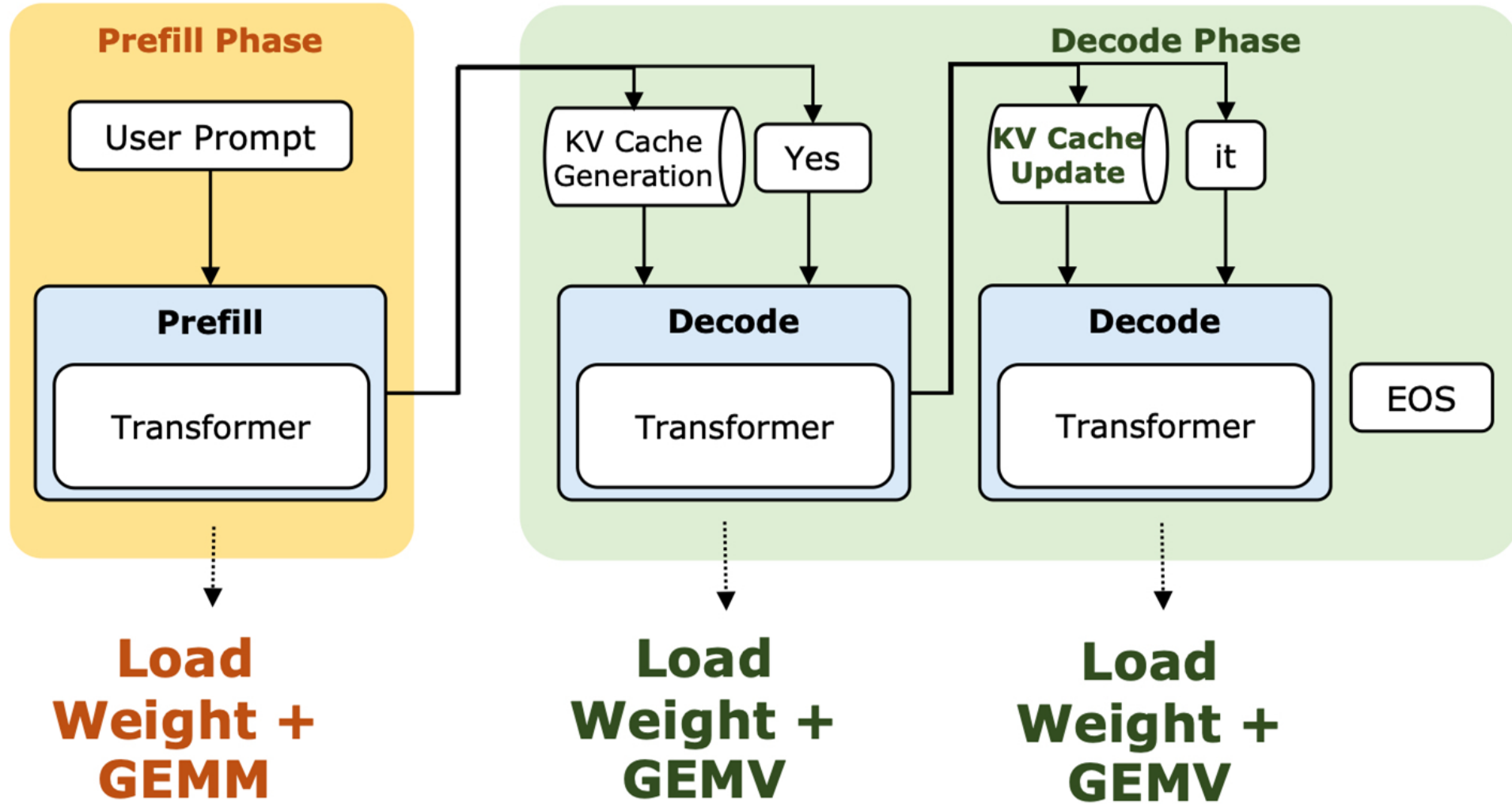
Need a very Large Model



Need compute and acceleration



Inference



Bottlenecks for LLM Acceleration

- Memory bandwidth
- Interconnect latency
- Model-parallel overhead
- Precision and quantization
- Cost / Energy

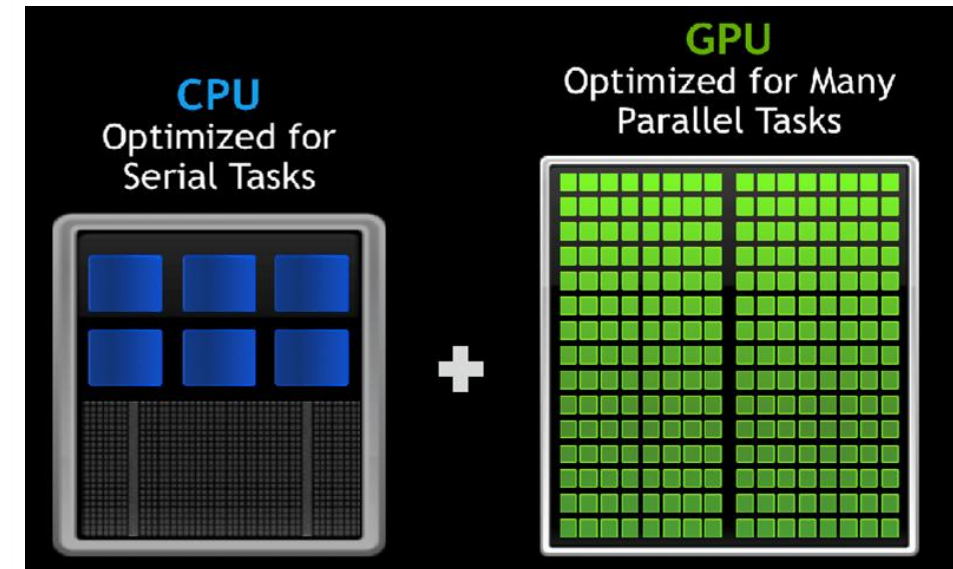


Algorithm-level acceleration

Generating one word at a time is slow



How to better exploit GPU in parallel processing?



Speculative decoding

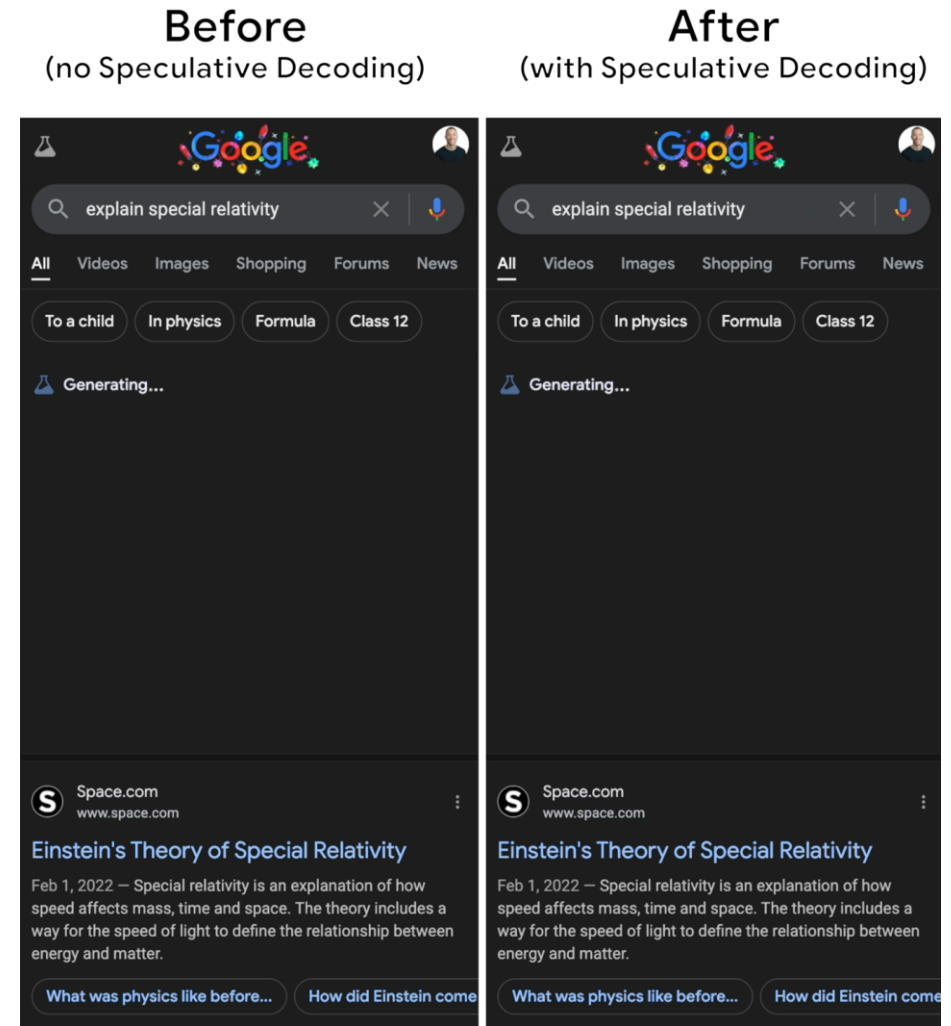
A technique that can reduce the inference times for LLMs significantly

The larger the LLM, the more competent it can be. However, these larger models are also slower, because, for example, each decoding step needs to read the entirety of the model's weights. This can mean that the **model needs to read on the order of a tera-byte of data for each word it produces**

An LLM generates its output one token at a time (*autoregressive*), where a token is usually a word or a part of a word.

-- a technique that can reduce the inference times for LLMs significantly

Google is using it →
Much faster result generation



Motivation 1: Some tokens are easier to generate than others

What is the square root of 7? The square root of **7** is **2.646**

- Generating the emphasized token “**7**” is relatively easy; for example, we can notice that the previous tokens “square root of” happened before, and just copy the following token
- Generating the tokens “**2.646**” is harder; the model would need to either compute or remember the answer.

This observation suggests that the large models are better due to better performance in difficult cases (e.g. “**2.646**”), but that in the numerous easy cases (e.g., “**7**”), small models might provide reasonable approximations for the large models.



Motivation 2: The bottleneck for LLM inference is usually memory

Machine learning hardware varieties, TPUs and GPUs, are highly parallel machines, usually capable of *hundreds of trillions* of operations per second, while their memory bandwidth is usually around just *trillions* of bytes per second (a couple of orders of magnitude lower)

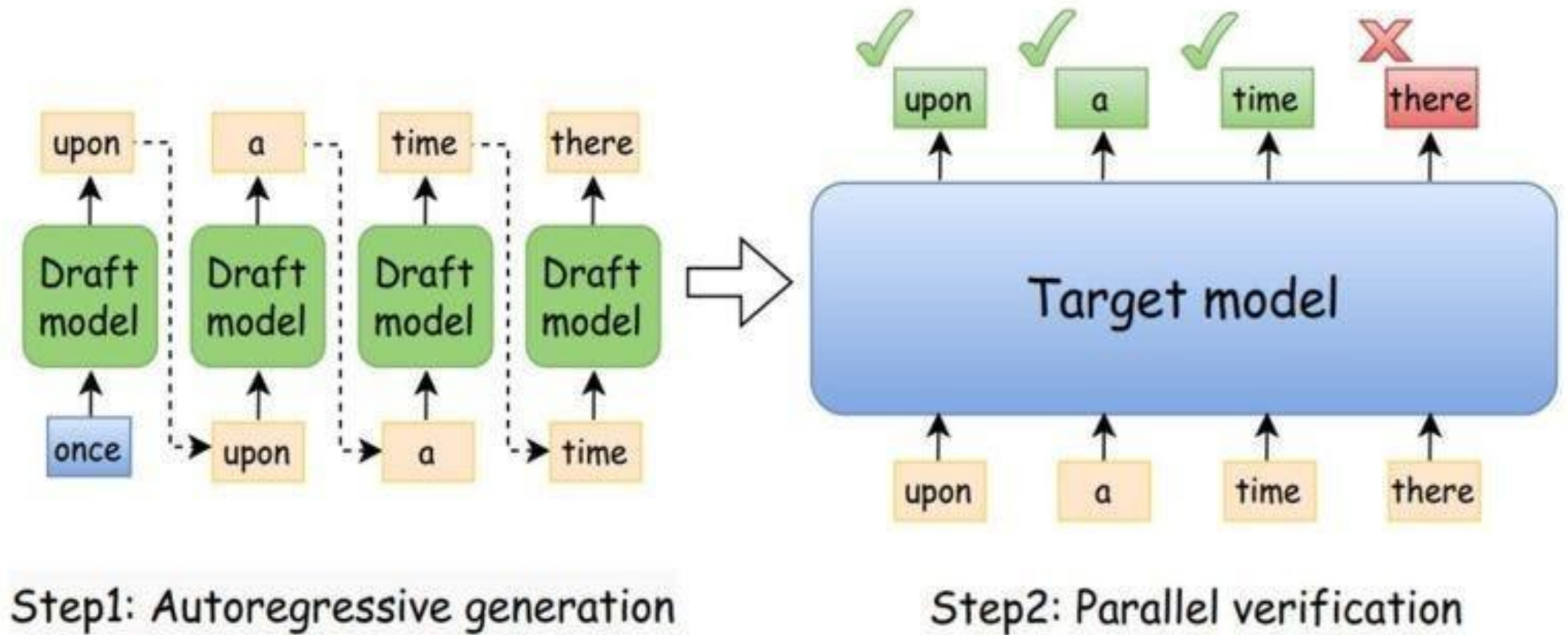
Hardware can do	Transformers need
~100s–1000s operations/byte read	~10 operations/byte read

This means that when using modern hardware, we can usually perform hundreds of operations for every byte read from memory.

In contrast, the Transformer architecture that underlies modern LLMs usually performs only a few operations for every byte read during inference, meaning that there are ample spare computational resources available when generating outputs from LLMs on modern hardware.



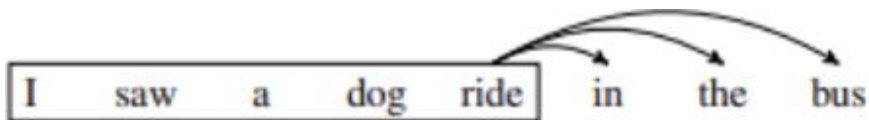
Speculative Decoding overview



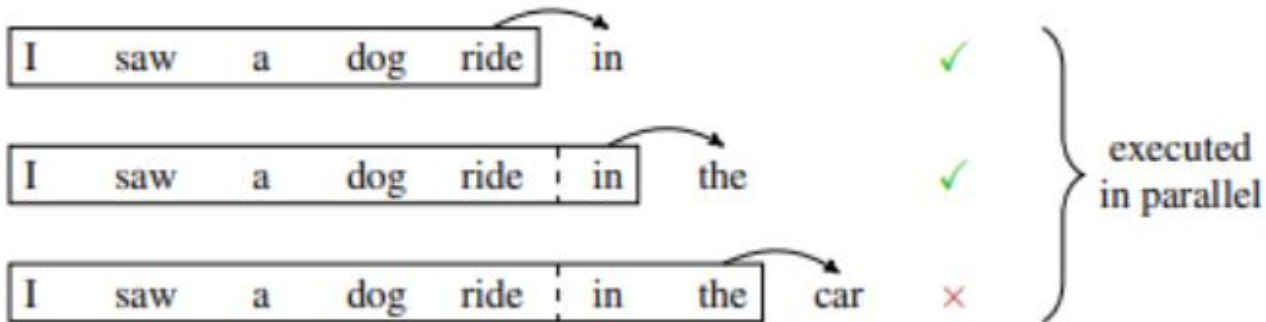
Speculative Decoding: An example

Small amount of time due to using a smaller model

Predict



Verify



Use Large Models, but verify in parallel

Accept



Take the accepted words



Generative AI



My preferred taxonomy

Generative AI

**BIG DATA(PROVIDER'S and YOUR) + Statistics
+ Neural networks + Optimization**

(4)

ChatBot

Drafting content

Writing code

Generating image, video

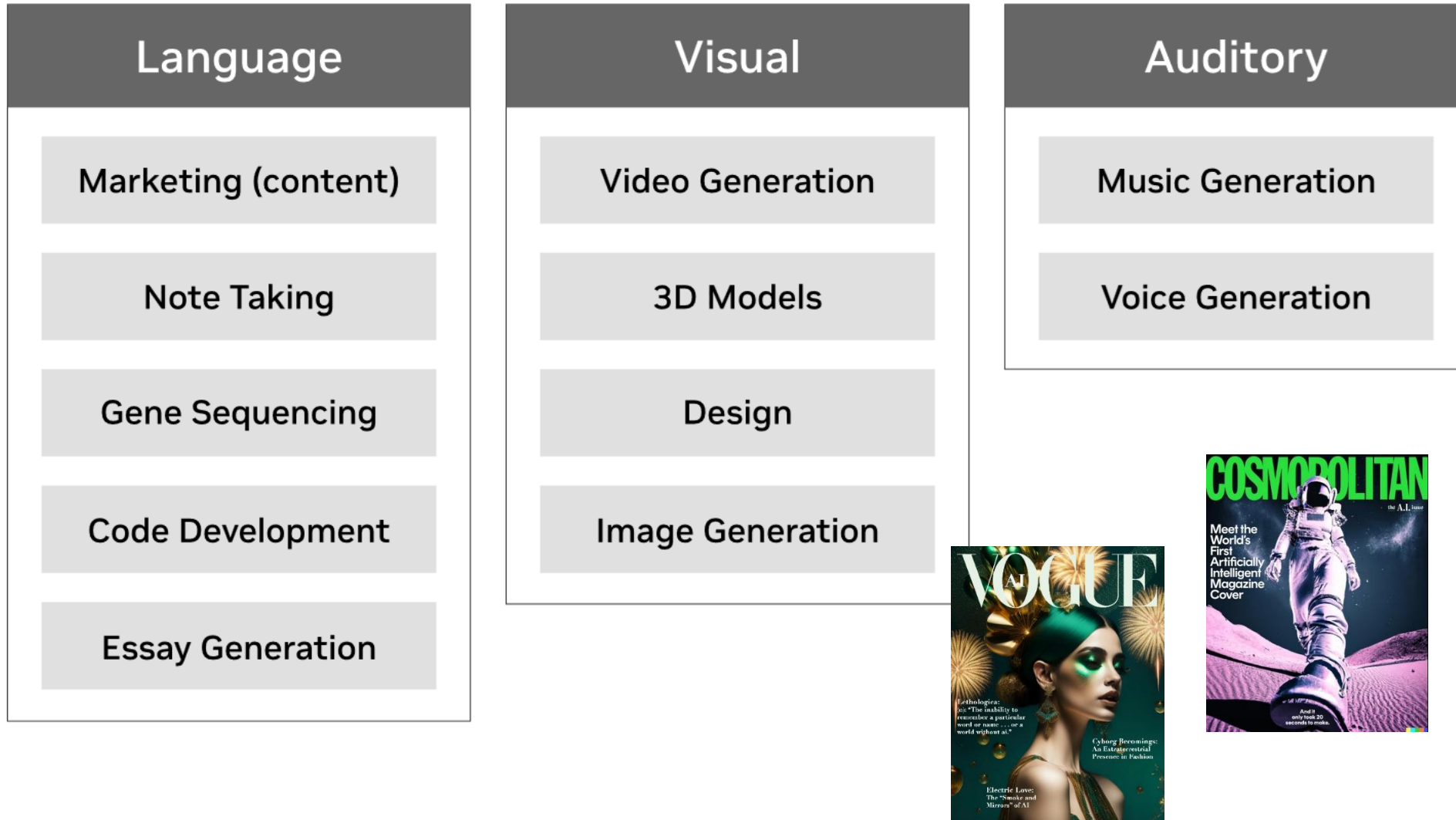
...

(5) AI agent

(6) Agentic AI



Generative AI applications



Where cannot we really evaluate?

GenAI has great advantages in creative fields

- Visual art
- Music
- Poetry
- ...

In art there is no true/false, right/wrong,
just like/dislike. So, all is debatable



GenAI for text generation, opinions, summaries is based on statistics
(*probabilities never reach 100%*)

***Sometimes they work. Sometimes don't, but we do
not have real number to evaluate the quality
In some instances, evaluation is a personal taste***



Generative AI: How does it work?

Generative AI models use **deep learning** to identify the patterns and structures within existing data to generate new content in a **statistical way**. It can leverage different learning approaches, including unsupervised or semi-supervised learning

Generative AI models is trained on **massive datasets** through **huge neural networks**, e.g.,

- OpenAI trained GPT-3 on 500 billion tokens/words; 13,000 billion tokens of Web text, code, and other data sources were used for training GPT-4 (actually, 6,000 billion tokens processed over multiple epochs)
- Trillions tokens were used for Gemini 3 Pro trained on Google's TPU v5e and v6e pods

GenAI learns patterns and can generate new content that **mimics human creativity**

- It is a valuable tool for many industries such as gaming, product design, art creation, entertainment, documentation, etc.
- It can even assist in scientific research as in material and drug production, design clinical trials, etc.



LLM and RAG

- **Large Language Models (LLMs)** are the foundation: they are massive neural networks trained to understand and generate human language.
 - They take an input and produce a response based on patterns learned during training
 - Powerful, but limited to what they already know
- **Retrieval Augmented Generation (RAG)** connects an LLM to external information sources like documents or databases
 - Before answering, the system stores and then retrieves through **vector databases** relevant content and uses it to respond
 - This reduces errors and keeps answers up to date, but **introduces real engineering work around embeddings, vector search, and document design**



Vector databases and Vector search

- Embedding models trained on massive datasets create for each “resource” a **vector of hundreds or thousand of mathematical parameters** (*dimensions*) that can capture the characteristics of the data in input
- How to retrieve data quickly and semantically?
- **Vector indexing** → *approximate nearest-neighbor* → they do not aim to find the exact correspondence (like in relational databases) but something that is **close enough**
 - HSNW
 - IVF
- **Vector database**
- **Vector search** is an AI-powered information retrieval technique (*similarity search*) that locates data based on **conceptual similarity** rather than *exact keyword matching*. It converts data (text, images, audio) into numerical vector embeddings using machine learning, allowing systems to find "nearest neighbors" in a high-dimensional space



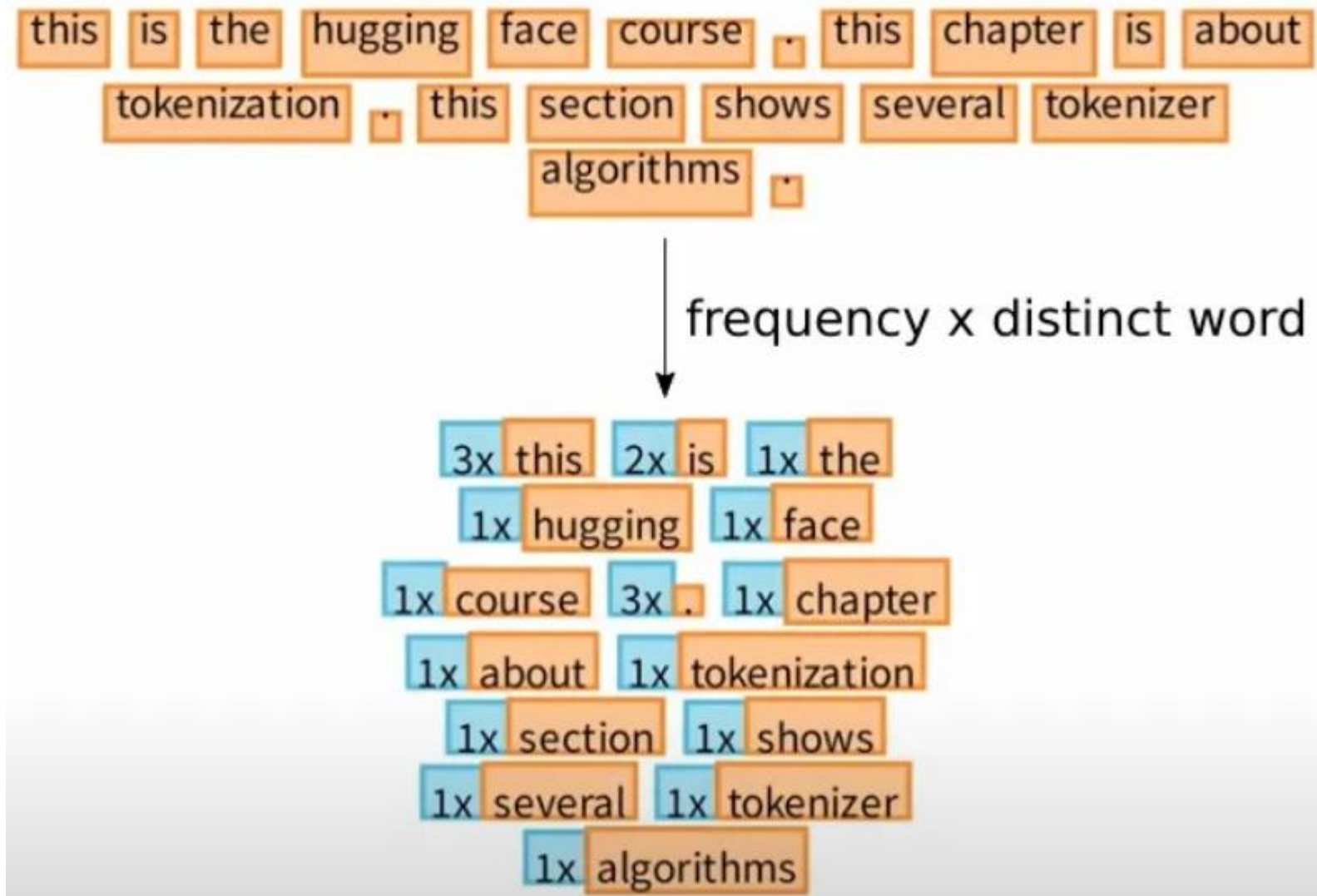
Subword tokenization algorithm

- Training a tokenizer is a statistical process that tries to identify which subwords are the best to pick for a given corpus, and the exact rules used to pick them depend on the tokenization algorithm, such as **Byte-Pair Encoding (BPE)**
- BPE is a *sub-word* tokenization algorithm that, after training, **breaks text into smaller pieces** called **tokens**. For example: “hugs” → (trained) BPE → “hug”, “s”
- BPE was originally invented as a *text compression algorithm*. It was later adopted in the context of machine translation first in *Natural Language Processing (NLP)*, and then *used by OpenAI for tokenization during the pretraining the GPT model*, as well as many other Transformer models
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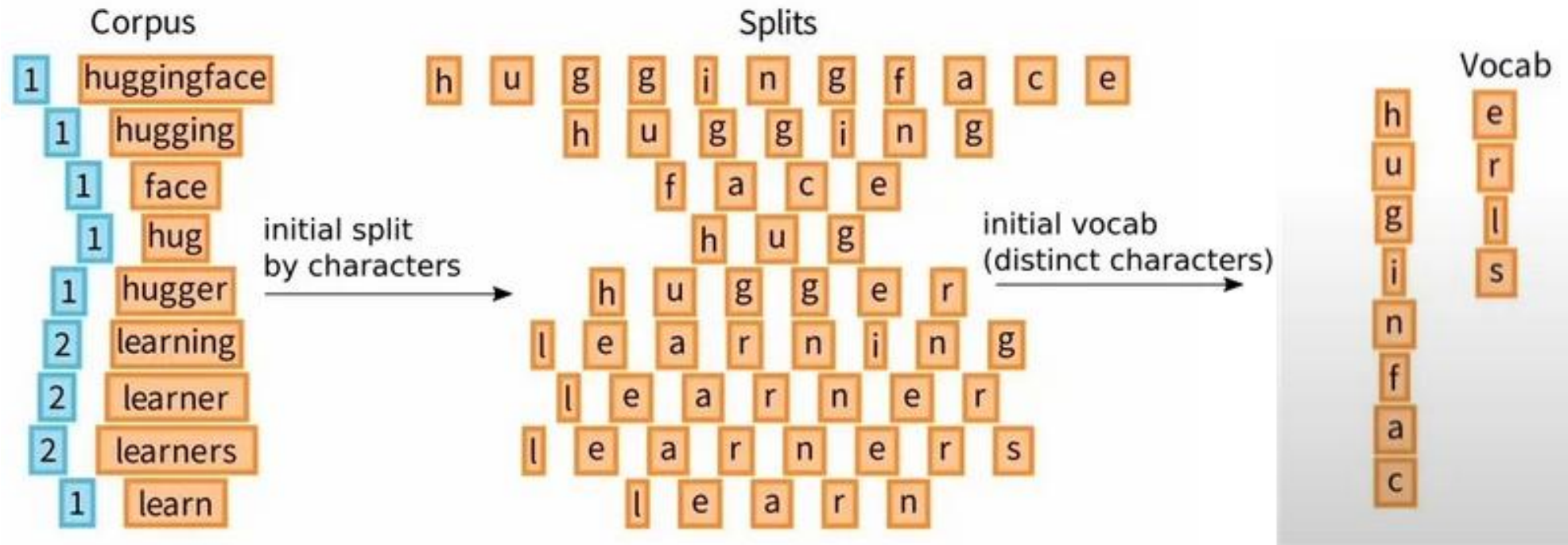
BPE training

BPE training starts with a standardized and pre-tokenized corpus. As an example, consider the following raw corpus that can be represented as the convention ***frequency x distinct*** word instead, which we will follow throughout the rest of this lesson



Initial splits and vocabulary

Given a corpus following the above-mentioned convention, as illustrated in the figure below, we first split each distinct word into characters. The set of distinct characters that appear in these words forms the initial vocabulary (*vocab*). For example, the vocab shown below contains 12 characters →



Expanding the vocab

- **BPE training starts with the initial vocabulary and gradually expands it to reach the desired vocabulary size, denoted as n .** Let us say we want to have a vocabulary size of $n = 20$
- Training proceeds by repeatedly performing the following steps:
 - Calculate pair frequencies, by sliding over each pair of consecutive characters in the current splits
 - Choose the most frequent pair from this set; this becomes the next merging rule
 - Update the splits and vocab by applying the new merging rule



count each pair frequency
by sliding window (2 consecutive tokens)

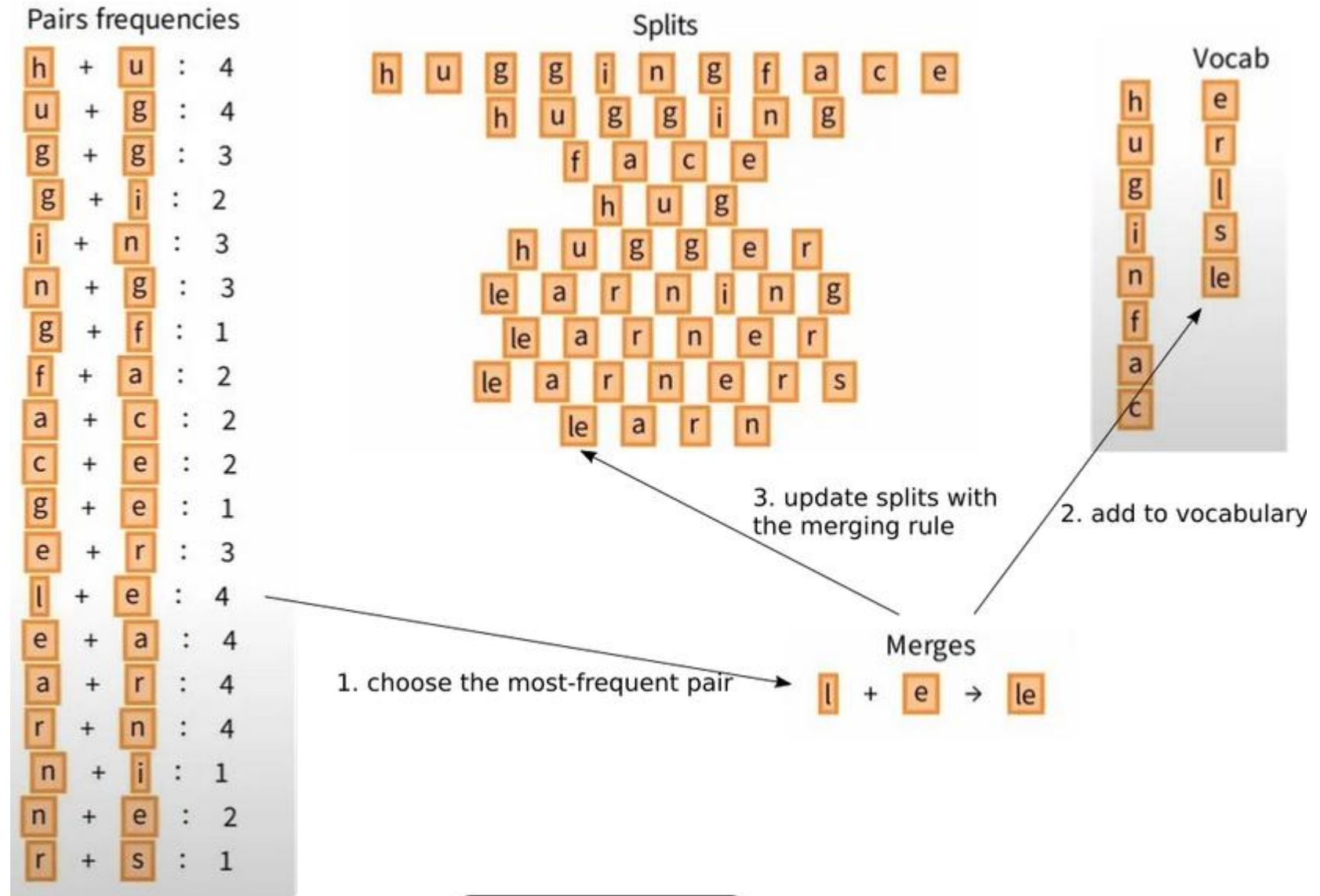
Pairs frequencies

h	+	u	:	4
u	+	g	:	4
g	+	g	:	3
g	+	i	:	2
i	+	n	:	3
n	+	g	:	3
g	+	f	:	1
f	+	a	:	2
a	+	c	:	2
c	+	e	:	2
g	+	e	:	1
e	+	r	:	3
l	+	e	:	4
e	+	a	:	4
a	+	r	:	4
r	+	n	:	4
n	+	i	:	1
n	+	e	:	2
r	+	s	:	1



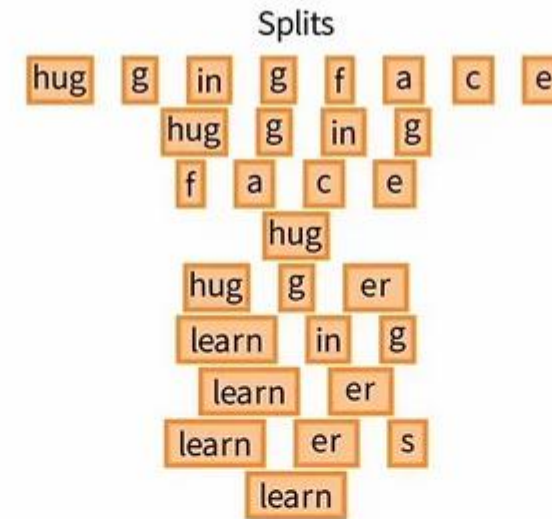
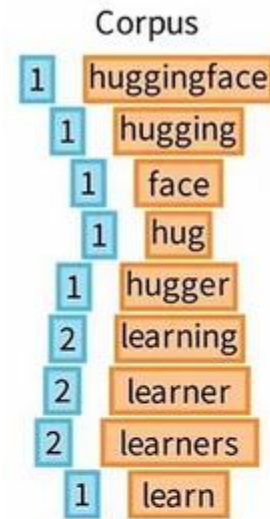
Continuation of the iterative process

This iterative process continues until the vocabulary reaches the desired size, e.g., $n = 20$ as shown in the image below



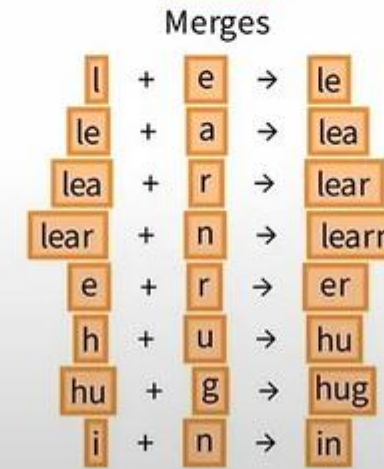
Continuation of the iterative process for training

That's it for BPE training!
Now we can apply our
(trained) BPE to new text to
break it into smaller
subword units/tokens.



Pairs frequencies

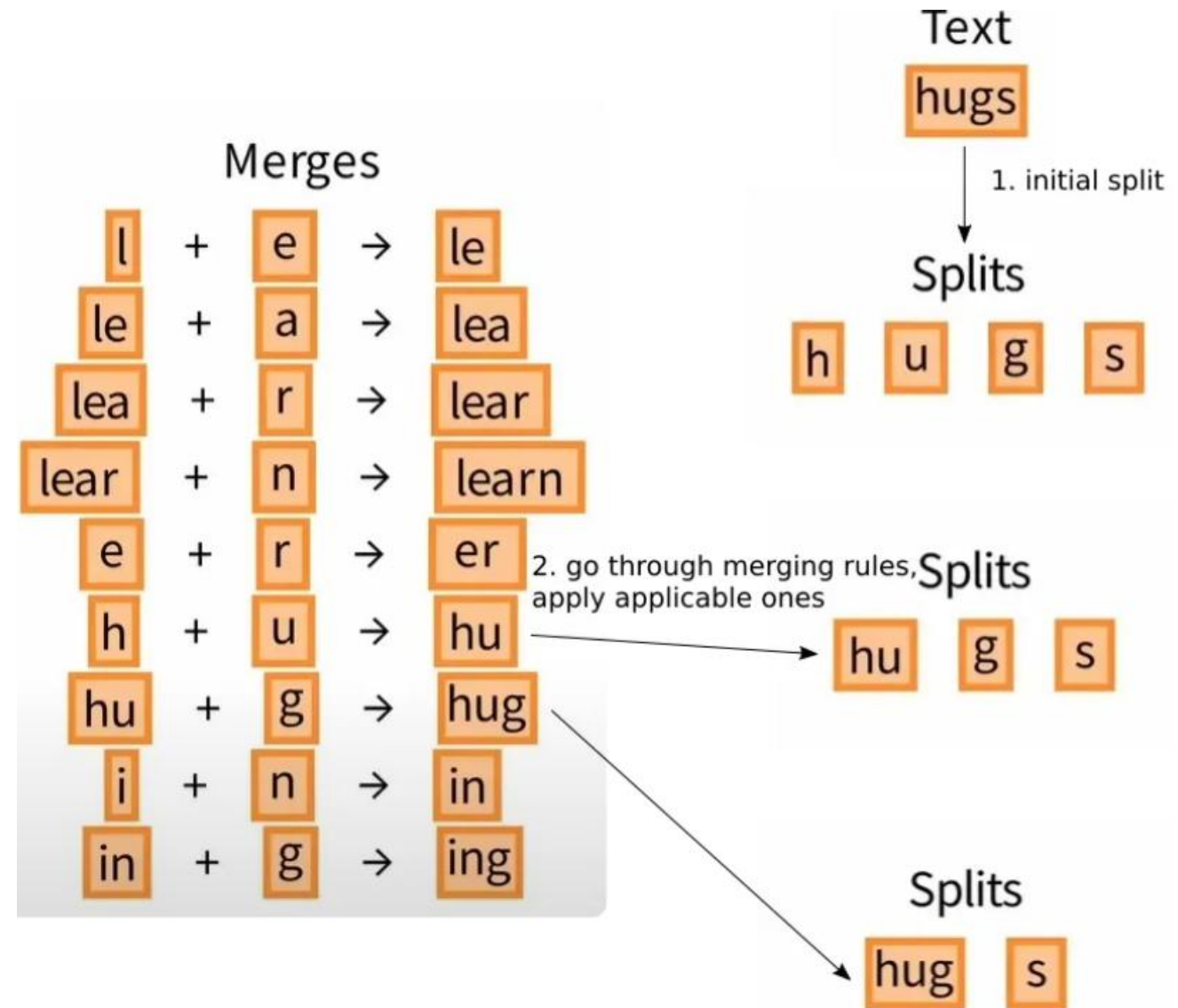
hug	+	g	:	3
g	+	in	:	2
in	+	g	:	4
g	+	f	:	1
f	+	a	:	2
a	+	c	:	2
c	+	e	:	2
g	+	er	:	1
learn	+	in	:	2
learn	+	er	:	4
er	+	s	:	2



Application of the derived BPE to new text

We start by splitting the text into individual characters:

- Initial split: 'h', 'u', 'g', 's'
- We then apply the learned BPE merging rules sequentially:
 - Applying the first merging rule ($h + u = hu$): 'hu', 'g', 's'
 - Applying the second merging rule ($hu + g = hug$): 'hug', 's'
 - At this point, no further merging rules apply, so we are finished.
 - Final tokenized output: 'hug', 's'



Video explanation

- We explored what BPE is and how it works, using on a text example
- The explanation is based on the video from *HuggingFace* that breaks down the concept of BPE in a visual way. Highly recommended if you prefer learning through video over text...

<https://www.youtube.com/watch?v=HEikzVL-IZU>

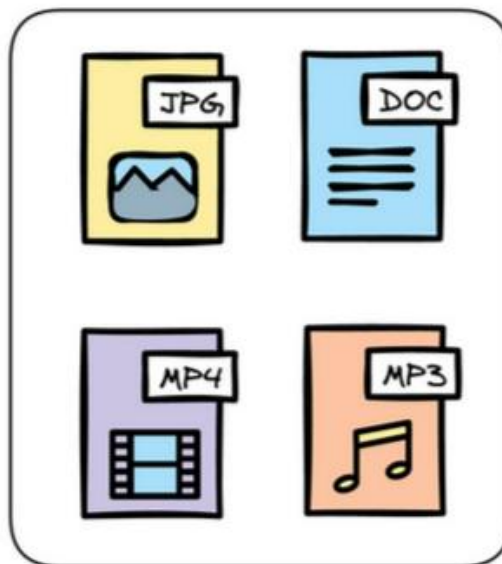


Vector database

A vector database is a specialized database designed to store, manage, and query high-dimensional vector embeddings that is a **numerical representations** of unstructured data like text, images, and audio

Unlike traditional databases that use exact matches, vector databases use **approximate nearest neighbor algorithms** to perform *semantic*, similarity-based searches, making them crucial for RAG-based AI applications

Unstructured data



0.2	-1.7	...	2.3
0.4	0.5	...	-1.7
4.1	-1.9	...	-1.5
-1.1	0.7	...	5.3
-3.5	2.3	...	0.5
-1.7	0.4	...	0.2
2.3	0.2	...	0.7
-1.9	4.1	...	-2.4
0.5	-1.5	...	2.3

Embeddings



Example

INPUT DATA (IMAGES)

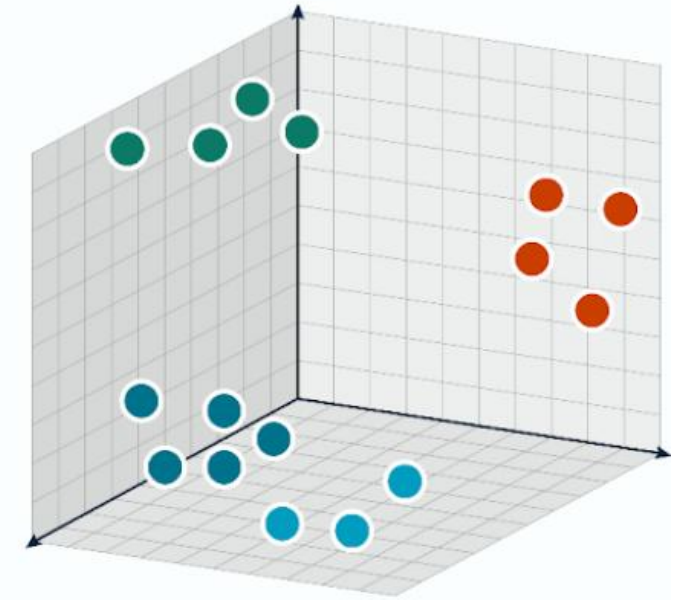


VECTOR EMBEDDING

```
[0.95, 0.49, 0.67, 0.75, 0.43, 0.00, 0.60, 0.13, 0.72, ...],  
[0.73, 0.83, 0.80, 0.48, 0.55, 0.83, 0.86, 0.85, 0.06, ...],  
[0.03, 0.71, 0.62, 0.07, 0.74, 0.30, 0.80, 0.08, 0.63, ...],  
[0.49, 0.48, 0.38, 0.51, 0.45, 0.05, 0.10, 0.92, 0.29, ...],  
[0.84, 0.38, 0.81, 0.00, 0.48, 0.61, 0.41, 0.85, 0.93, ...],  
[0.24, 0.90, 0.69, 0.57, 0.43, 0.60, 0.37, 0.63, 0.28, ...],  
[0.72, 0.33, 0.36, 0.98, 0.88, 0.05, 0.64, 0.91, 0.15, ...],  
[0.89, 0.17, 0.44, 0.78, 0.33, 0.59, 0.29, 0.23, 0.22, ...],  
[0.88, 0.75, 0.38, 0.77, 0.31, 0.98, 0.84, 0.79, 0.25, ...],  
[0.11, 0.77, 0.30, 0.56, 0.62, 0.14, 0.89, 0.19, 0.62, ...],  
[0.63, 0.41, 0.97, 0.73, 0.64, 0.41, 0.96, 0.36, 0.87, ...],  
[0.51, 0.16, 0.26, 0.94, 0.53, 0.38, 0.39, 0.90, 0.55, ...],  
[0.91, 0.99, 0.10, 0.87, 0.24, 0.91, 0.94, 0.36, 0.69, ...],  
[0.89, 0.64, 0.14, 0.94, 0.51, 0.24, 0.61, 0.99, 0.11, ...],  
[0.77, 0.65, 0.48, 0.94, 0.49, 1.00, 0.24, 0.10, 0.97, ...],  
[0.58, 0.47, 0.01, 0.56, 0.59, 0.32, 0.79, 0.95, 0.40, ...]
```

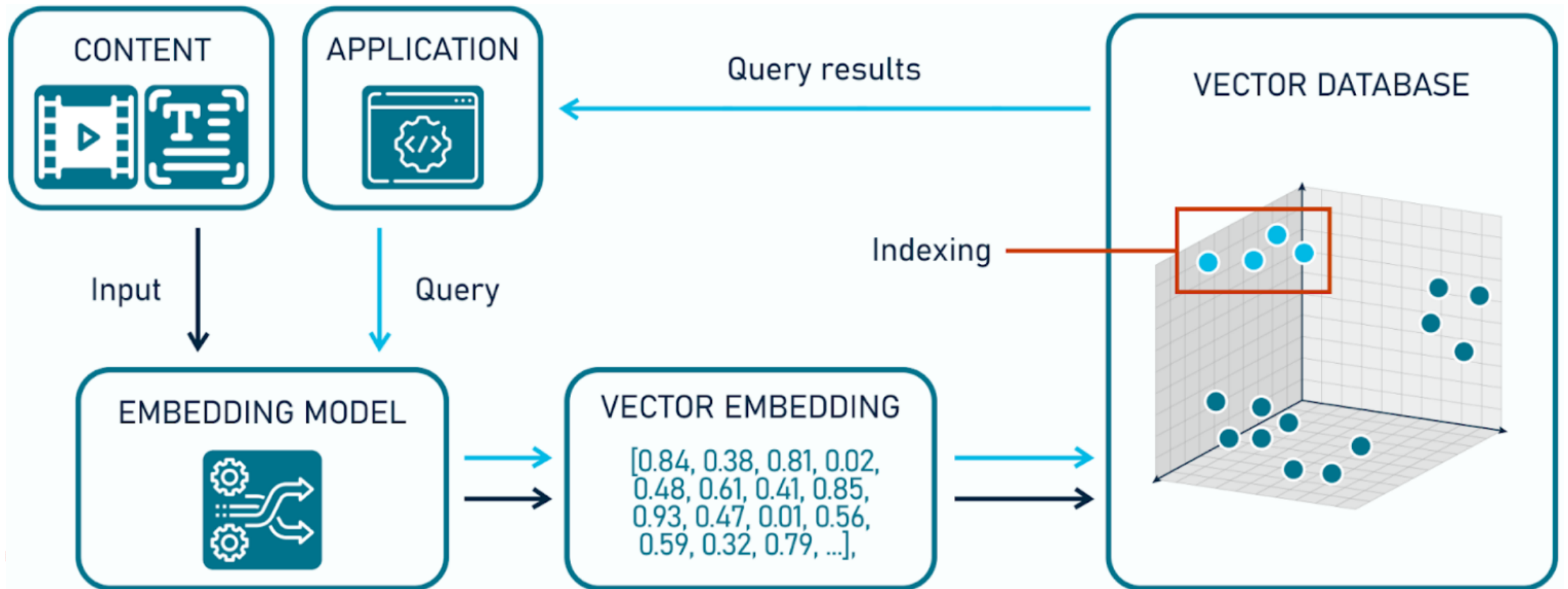


VECTOR DATABASE



Indexing

- ZCcscsczc



Generative AI applications

Generative AI has given organizations the ability to leverage a large amount of unlabeled data to create foundation models that can perform multiple tasks:

- **GPT-3** and **GPT-4**: texts, answers, essays, summaries, ...
- **Stable Diffusion**: generate photorealistic images given a text input
- **OpenAI SORA, Nano Banana**: video generation
- **Google MusicFX, Adobe Music GenAI Control, Suno**: music
- **Copilot, Cursor, Claude code**: generation of software
 - Anthropic and OpenAI engineers say **AI now writes 100% of their code**
 - **Graphite** is valid for testing code



Well-known intrinsic issues

- Explaining the results of a model with billions or trillions parameters, or understanding how it arrived at any given decision, is impossible
- **No company has disclosed the training data used to train the models** → many concerns about potential privacy violations or copyright infringement
- Current generative AI models have well-publicized limitations, such as “**hallucinations**” (generating glaringly false information) and “**biases**” that create new types of risk
- The models are beginning to demonstrate “emergent properties” that they were not specifically trained to replicate. From a practical perspective, this means that models intended for one function can (or will) be repurposed and used for other tasks



Why *hallucinations* are unavoidable (1/4)

- ***Hallucinations*** are a fundamental byproduct of how these models are designed, trained, and evaluated, rather than just a temporary ***bug***
- Hallucinations are **mathematically inevitable** due to the tension between the need to produce coherent, fluent language and the inability to guarantee factual accuracy in every scenario

GenAI is a tool

Use GenAI in the fields that you know to the extent that you can control the outcome



Why *hallucinations* are unavoidable (2/4)

1. Probabilistic nature

- **Next-token prediction:** LLMs are designed to predict the most statistically likely next word in a sequence based on patterns learned during training, not to retrieve verified facts from a database
- **Plausibility over truth:** When a model is uncertain, it is optimized to generate a plausible-sounding response rather than to admit it does not know. This “confident, wrong” behavior is a direct result of being **designed to create, not to verify**

2. Inherent training data and knowledge gaps

- **The Long Tail of knowledge:** No training dataset is complete. When asked about niche, rare, or new information (the "long tail" of data), the model lacks sufficient examples to learn accurate patterns and will "fill in the blanks" with plausible fabrications.
- **Incorrect data:** Models are trained on the Web, which contains false information, biases, and contradictions, which the model may confidently reproduce



Why *hallucinations* are unavoidable (3/4)

3. Evaluation Incentives (“The school-test problem”)

- **Reward Structure:** Current AI evaluation methods often punish uncertainty (“I don't know”) and reward guessing. Models are essentially trained to be “eager-to-please” interns that would rather make up an answer than admit ignorance
- **Accuracy-only metrics:** Benchmarks that only count “right vs. wrong” encourage models to take risks, leading to a higher rate of hallucinations

4. Mathematical and computational constraints

- **Information density:** A 2025 OpenAI study suggests that the total error rate in generating text is bound by the difficulty of distinguishing truth from falsehood
- **Impossible tasks:** Some tasks, such as accurately counting characters in a word or solving certain logical puzzles, exceed the structural capabilities of current transformer architectures, leading to systematic, unavoidable errors



Why *hallucinations* are unavoidable (4/4)

5. The “open world” vs. “closed world” issue

- **Unbounded environment:** Under the “open world” assumption (where the AI faces unpredictable, new inputs), hallucinations are unavoidable because no finite training data can guarantee accuracy in an infinite future
- **Contextual misinterpretation:** Even if data is accurate, the model may misinterpret the context, leading to incoherent or incorrect reasoning

Techniques like *Retrieval-Augmented Generation* (RAG) and better fine-tuning can *reduce* hallucinations, but **they cannot eliminate them completely** because they are a fundamental, intrinsic feature of current probabilistic AI, as validated in modern research



Agentic AI



What an agent is NOT

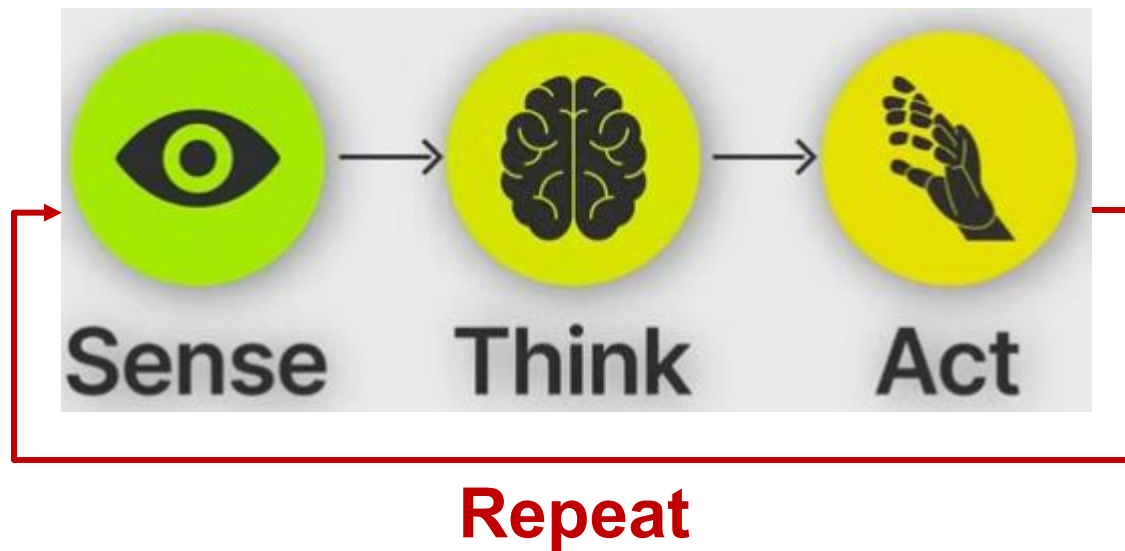
~~Chatbots~~

~~Prompt chains~~

~~Simple automations~~



Agentic AI



An autonomous system consists of:

- 1) **Reasoning model** (LLM) – The brain (e.g., ChatGPT, Claude, Gemini)
- 2) **Sensors** – Eyes and Ears (perception from any kind of digital source)
- 3) **Memory** – Store the acquired content
- 4) **Actuators** – Tools that can write, send a note, update data in a Database, connect to an API, etc.



Difference between *Automatic* and *Agentic* behavior

- **Automatic systems** proceed through a sequence of pre-established actions
- **Agentic systems** can self-evaluate the situation and change the sequence of actions (with some degree of autonomy). They can use technical frameworks, such as:
 - **Retrieval-Augmented Generation (RAG)**
 - **Re-Act**
 - **Multi-Agent System (MAS)**



System prompt

- Define output, e.g., format, structure,



Guardrails

- 1) Specify Do
- 2) Specify Don't

- 3) Add validation controls



- 4) Limit agent's tools to what it is needed and how agents can use them



Use prompt template

Search previous nodes' fields

alarm and affected tourism in Japan. \n 2. Sean "Diddy" Combs being acquitted of sex trafficking and racketeering charges but encountering a new denial on a separate issue. This event was reported from San Francisco on July 6, 2025. \n 3. Both briefs included a fun fact about the origin of the term "trivia." The fact explained that "trivia" originally meant information of little value, deriving from the Latin word "triviale..."

A refusal [null]

Annotations

A logprobs [null]

A finish_reason stop

Schedule Trigger 1 item

A timestamp 2025-07-04T15:58:23.156+02:00

A Readable date July 4th 2025, 3:58:23 pm

A Readable time 3:58:23 pm

A Day of week Friday

A Year 2025

A Month July

Expression

Anything inside {{ }} is JavaScript. [Learn more](#)

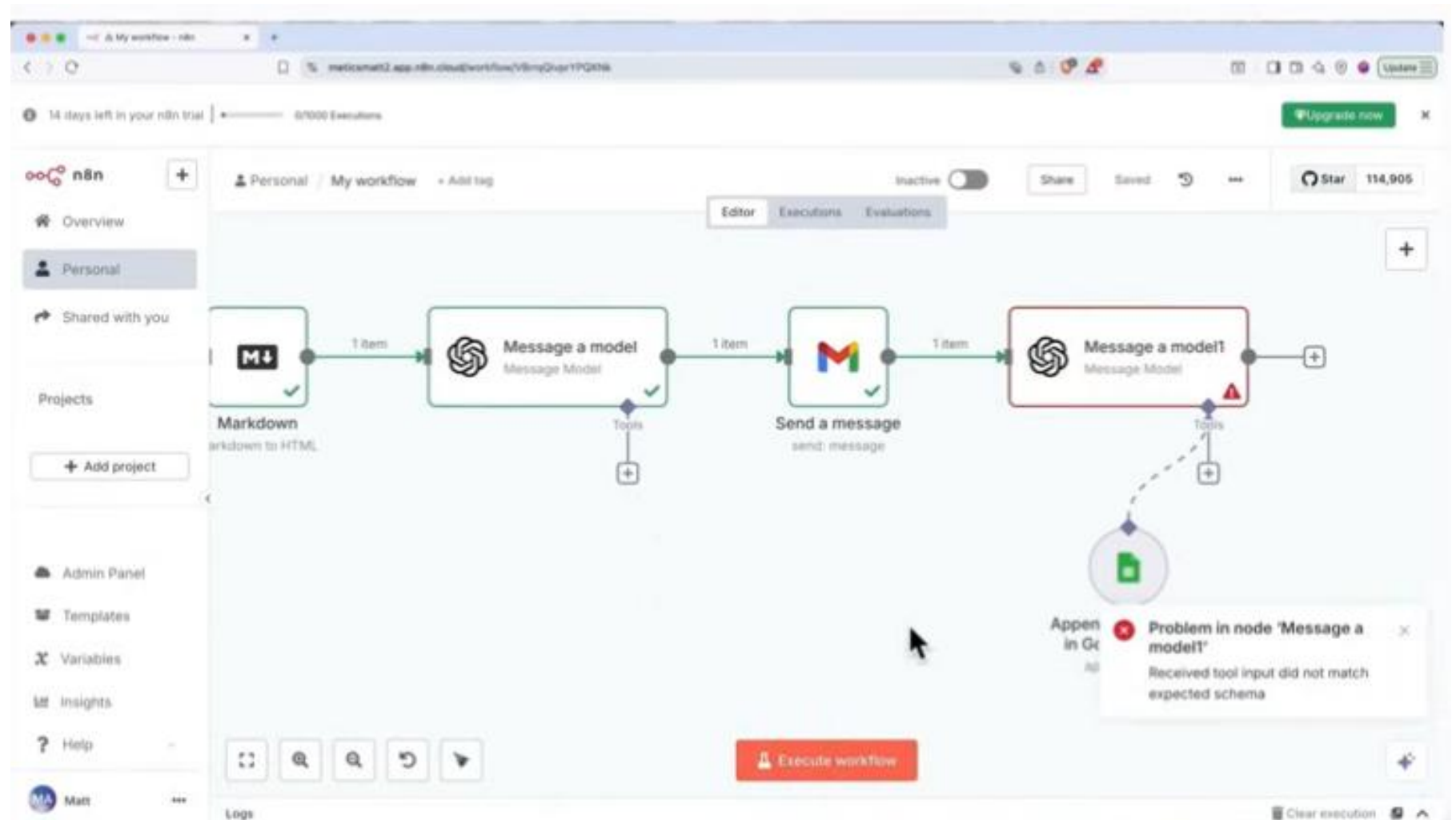
Every morning, gather the weather for San Francisco, pull in 2-3 positive, trending news stories, and find a fun fact from Wikipedia. Don't repeat anything from previous days. Format it like a mini email newsletter in Markdown. Avoid the following topics:
{{ \$json.message.content }}
Today's date: {{ \$('Schedule Trigger').item.json['Readable date'] }}

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Michele Colajanni - AI for SRS, 2026

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Expect some trials and errors



One agent per task

- Writing
- Formatting
- Sending
- Logging



Writing



Formatting

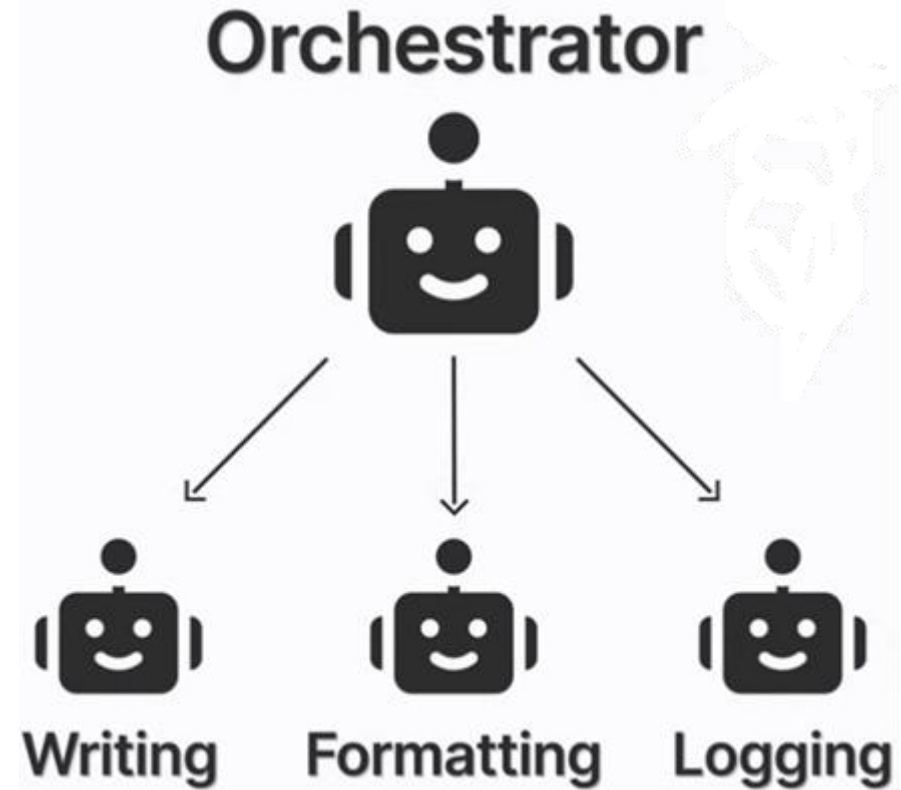


Logging



Orchestrator

•



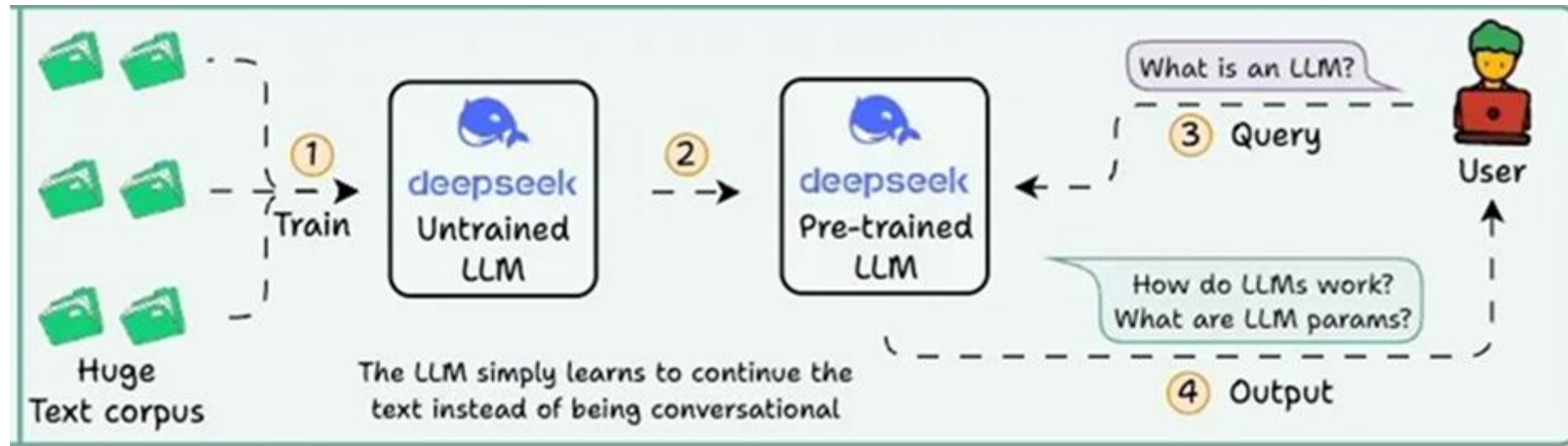
LLM training: Stage 1 of 5

Initialize the model: randomly sample from some distribution)



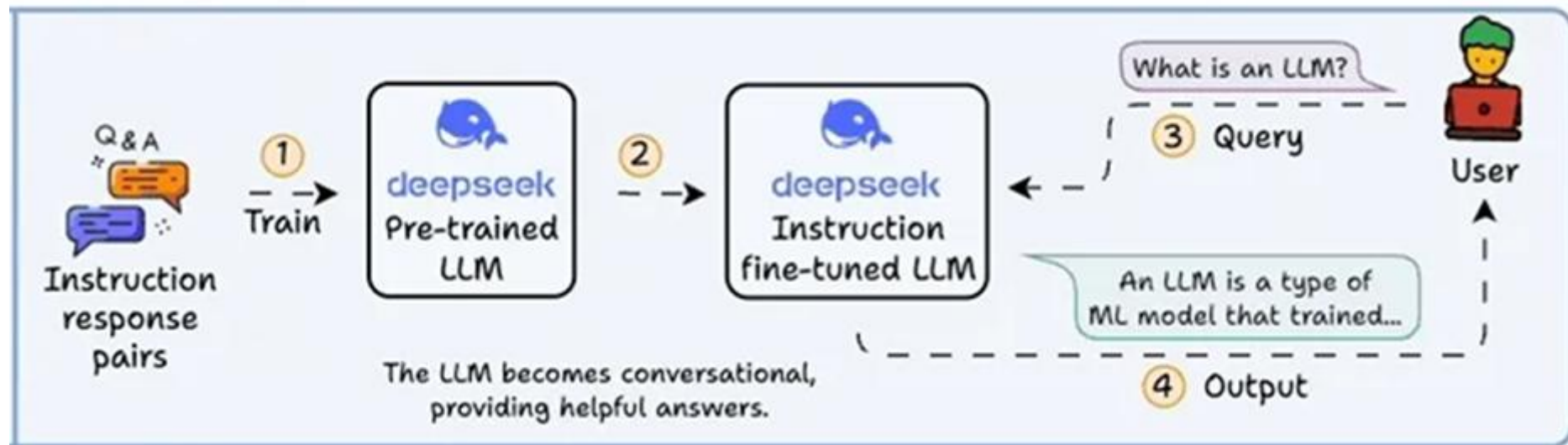
LLM training: Stage 2 of 5

Pre-training: Next Token Prediction



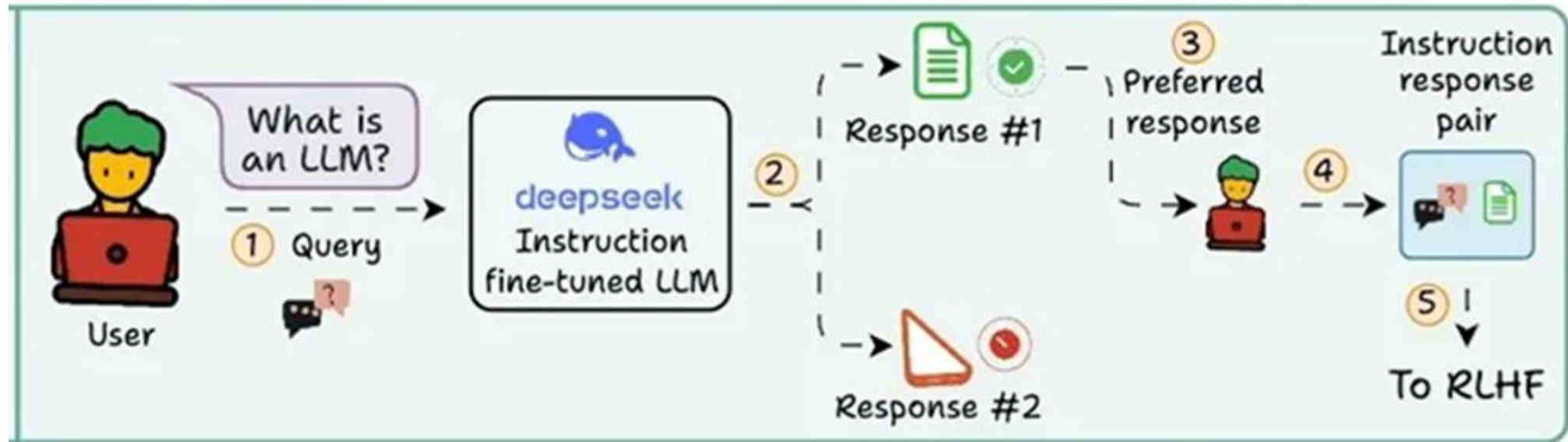
LLM training: Stage 3 of 5

Instruction fine-tuning: Making LLM conversational



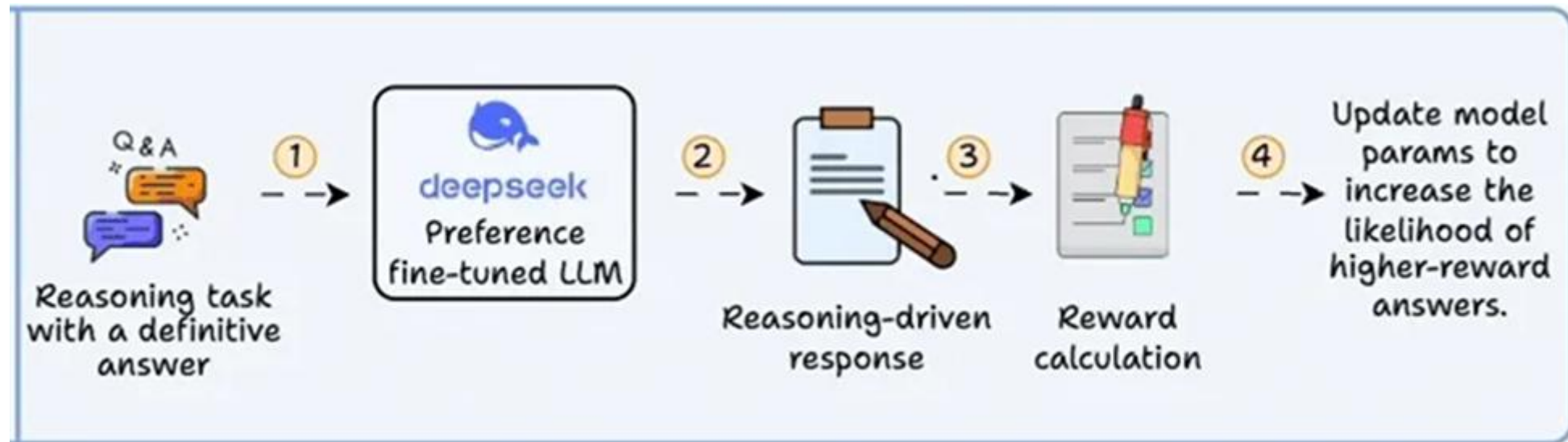
LLM training: Stage 4 of 5

Preference fine-tuning: aligning it with humans' values



LLM training: Stage 5 of 5

Reasoning fine-tuning: train models with specific reasoning structure and Reinforcement learning



Not everyone likes guardrails

- **DISAPPOINTMENT**

PENTAGON CHIEF **PETE HEGSETH** CALLED ANTHROPIC CEO **DARIO AMODEI** TO EXPRESS HIS “FRUSTRATION” OVER THE LIMITATION IMPOSED BY ANTHROPIC ON THE USE OF AI ON THE “CLAUDE” SOFTWARE THAT WAS USED BY THE US IN THE RAID IN VENEZUELA TO CAPTURE MADURO

- **MOTIVATION**

THE COMPANY DOES NOT WANT TO GIVE THE TRUMP ADMINISTRATION **FREE HANDS** ON THE **MASS SURVEILLANCE** OF AMERICANS AND THE **DEVELOPMENT OF “AUTONOMOUS WEAPONS”**, CAPABLE OF STRIKE WITHOUT HUMAN INTERVENTION

- **THREAT**

THE PENTAGON THREATENS TO TERMINATE ITS \$200M CONTRACT WITH ANTHROPIC



Encoder purpose

- **Read** and “understand” the input text
- **How:** Processes the entire input sequence(e.g., a sentence to be translated) and creates a context-rich of numerical representations
- **Output:** A set of vectors that represent the meaning of the sentence.



Decoder purpose

- **Generate** the output text
- **How:** Takes the representation of the encoder and, word by word, generates the output sequence (e.g., the translation)
- **Mechanism:** Pays attention to both the generated words and the input representation

Models like GPT are primarily based on Decoder-only architectures



Tokenization

- The entire text is broken into “tokens” (words or pieces of words)
- Common words are **whole tokens**, while others are broken
- Mathematical models do not understand words, only numbers
 - A unique numeric ID is assigned to each token
 - Intelligent algorithms (e.g., **BPE**) create a vocabulary of common tokens and sub-words to handle any text. **Byte-Pair Encoding (BPE)** is a data compression and tokenization algorithm that iteratively replaces the most frequently occurring pair of consecutive characters or bytes with a new, single token
 - Widely used in LLMs, BPE bridges character-level and word-level tokenization by building sub-word vocabularies. This method ensures rare words are broken down into recognizable, smaller units, allowing for an efficient, open-vocabulary approach that avoids unknown token issues



Embeddings

- Once we have the numeric IDs of the tokens, we need to associate them with a “meaning”
- **An embedding is a vector** (a long list of numbers, e.g., 4096) **that represents the meaning of a token**
 - These vectors are learned during the pre-training phase
 - The model learns to place the tokens in a multidimensional “semantic space” based on their use in the text, e.g.,
 - **Proximity = Similarity**: Cat and Dog are close. Cat and Car are far away
 - **Direction = Relation**: The route from Italy to Rome is similar to that from France to Paris (the relation is "country-capital")
 - Relationships between words (e.g., “King” is to “Queen” as “Man” is to “Woman”) can become mathematical operations between vectors
 - Thanks to the **attention mechanism** the embedding of a token can change based on the words surrounding it



The Fundamental Engine: *Predicting the Next Token*

- The heart of an LLM is to predict the most likely word that will follow a sequence of text
- Models are pre-trained on huge amounts of text from the Internet to learn these probabilities
 - "The cat sat on the..." → sofa (*high probability*)
 - "The cat sat on the..." → elevator (*low probability*)



Where does the “knowledge” of an LLM come from?

- **Data:** Hundreds of terabytes of text. Books, articles, code, conversations... almost the entire public Internet
- **Computation:** Thousands of GPUs running for months
- **Cost:** millions of dollars for one training cycle

This process is why only a few large companies with a lot of money can create “basic” models from scratch!



Key limitations and risks (1/2)

(LLMs are powerful, but they are not perfect. It is crucial to know their limitations)

Hallucinations: The model generates false, invented, or non-factual information, but presents it with great confidence

- Motivation: They are a **side effect of predicting the next token** → The model optimizes for statistical consistency, not for truth

Example

Asking for a biography of a nonexistent person and getting a (false) detailed one



Key limitations and risks (2/2)

Biases: The model reproduces and amplifies biases (social, gender, racial) present in the huge volumes of text on which it was trained

- Motivation: **The set of *training data* reflects the real world**, with all its biases

Example

Associating certain professions with specific genders



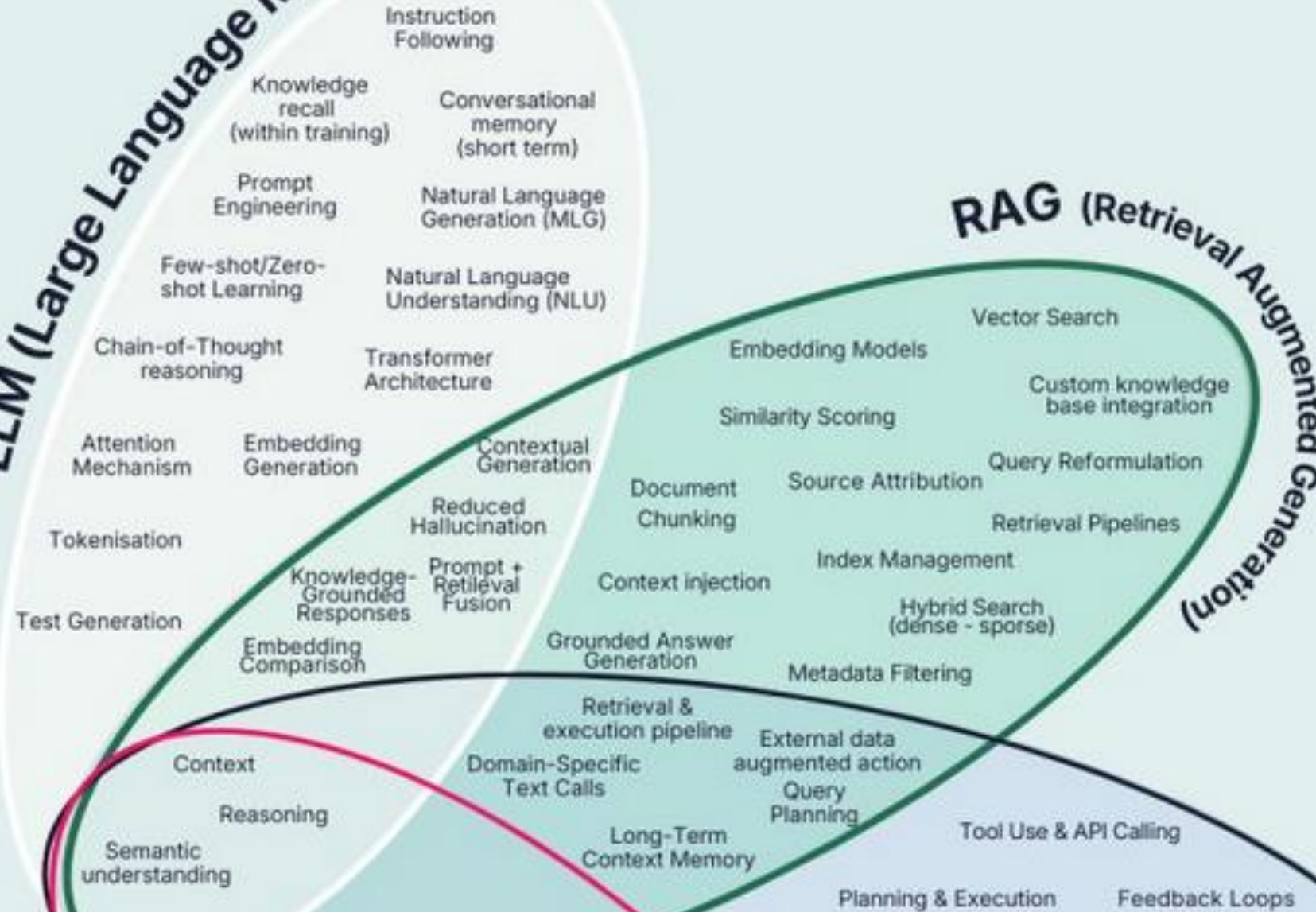
RAG



↑ Nuovi post

LLM (Large Language Model)

RAG (Retrieval Augmented Generation)



- You find layers:
 - Context
 - Memory
 - Retrieval
 - Tools
 - Autonomy



AI agents and Agentic AI

- **AI agents** go a step further
 - They are systems that can take actions on your behalf to achieve a task
 - They plan steps, call tools, execute actions, and track state over time
 - This is no longer just about answering questions. It is about **doing work to achieve a goal**
- **Agentic AI** systems push autonomy even further
 - They **coordinate** multiple agents, reason over longer horizons, manage memory, and pursue goals with limited human input
 - Here the challenge shifts from **building capability** to **orchestrating** behavior



Technology and Organizations

Organizations move up this stack technologically faster than they move organizationally

- **LLM** → *tooling* decision
- **RAG** → *data quality* decision
- **Agents** → *governance* decision
- **Agentic AI** → *coordination* and *accountability* decision

The technical maturity curve is clearer than the organizational one

The real constraint is not technical capability, but it is operating an organization with adequate model readiness



RAG PII protection (1/3)

Eliminating Personally Identifiable Information (PII) from Retrieval-Augmented Generation (RAG) documents is critical to preventing data leakage, ensuring GDPR/CCPA compliance, and protecting user privacy

The most effective approach is a defense-in-depth strategy, removing or masking PII during the ingestion (pre-vectorization) phase.

Before documents are converted into embeddings and stored in a vector database, they should be cleaned to remove sensitive data. Key strategies for eliminating PII from RAG:

1) **PII redaction and masking** (*Pre-Ingestion*)

- **Tokenization/Pseudonymization:** Replace PII (names, emails, SSNs) with consistent, irreversible tokens or placeholders (e.g., replacing "John Doe" with [NAME_1]). This preserves the context for the LLM without revealing the actual identity
- **Named Entity Recognition (NER):** Use automated tools like **Microsoft Presidio** or **Amazon Comprehend** to detect and redact sensitive entities
- **Regex-based Filtering:** Implement regular expressions for structured data like phone numbers, credit card numbers, or email addresses



2) In-flight masking (Retrieval & Generation)

- If PII cannot be removed at ingestion, it must be removed before being sent to the LLM.
- **PII Masker Tools:** Use tools like **PII Masker** (by HydroX AI) which can be integrated into vector databases like [Milvus](#) or [Zilliz](#) to mask data on the fly.
- **RAG-Level Filtering:** Apply sanitization to retrieved text chunks before they enter the LLM's context window.
- **Local LLM Pre-processing:** Use a smaller, locally hosted LLM to sanitize documents before passing them to a public LLM API.

3) Data-level security and synthesis

- **Synthetic data generation:** Replace real, sensitive documents with synthetic versions that retain the semantic structure and knowledge but contain no real PII.
- **Encryption at rest:** Ensure the vector database and underlying storage (e.g., S3 buckets) are encrypted, restricting access to authorized users only.
- **Data minimization:** Only ingest necessary information into the RAG system. If PII is not needed for the query, do not include it in the source files



4) Specialized tools and frameworks

- **Microsoft Presidio:** An open-source library for PII detection and anonymization
- **Guardrails AI:** Used to check and filter LLM responses for PII leaks
- **Eraser4RAG:** A technique designed to remove user-defined private knowledge from documents while maintaining public knowledge for generation
- **Data prep kit:** A tool designed to prepare documents, including removing sensitive data, for LLM applications

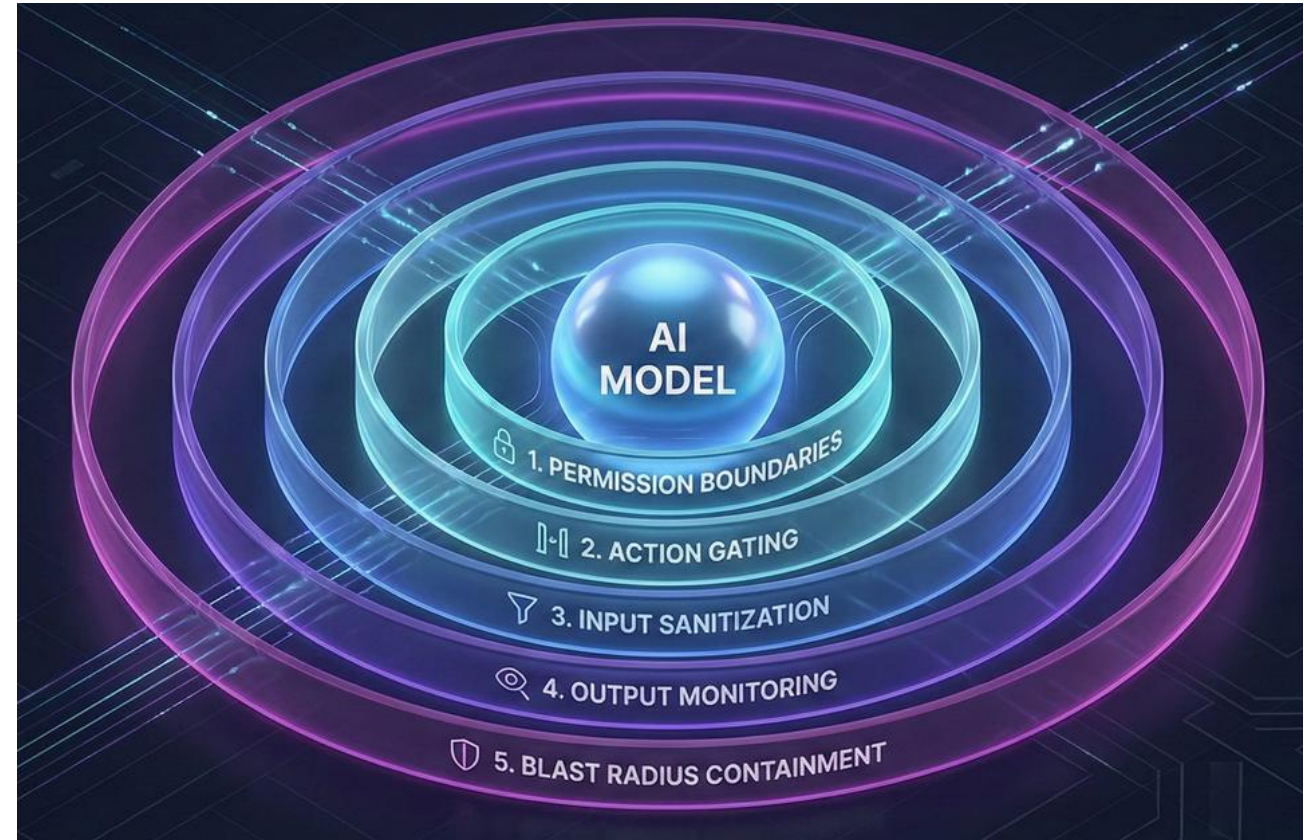
5) Best practices for RAG PII protection

- **Avoid raw data:** Never send raw, unprocessed PII to third-party LLM providers
- **Use no-retention APIs:** Use LLM APIs that explicitly state they do not use customer data for training
- **Audit regularly:** Conduct regular audits of your RAG pipeline to ensure no new PII has slipped through
- **Leverage metadata:** Tag sensitive documents with metadata for easier filtering and mgmt



Impact on organizations

- Organizational and business maturity is way harder and longer to gain than technical
- This occurs always with new technological revolutions
- To see less ROI-fails, it is convenient to start with some **small experiment-test-learning** view more instead of focusing immediately on real (challenging) projects



The long journey of organizations

- This photo captures the journey perfectly. Each step demands not only **technical capability** but also **stronger governance** and **leadership decisions**
- In the latest AI evolution, the question of accountability is becoming more important, especially in increasingly anonymous workflows
- **The real progression of AI is not autonomy. It is liability**
 - Every layer up the stack increases the blast radius. That is why “designing for trust” is the work with **measurable accuracy**, **reproducibility**, **authorization**, and **monitoring**. Without the integration of these features in your projects, confidence and outcomes will not scale



From AI to trusted AI

Each layer demands more than better technology

- Moving from LLMs to RAG means investing in **data quality** and **retrieval design**
- Moving from RAG to Agents means **trusting systems to act**
- Moving to Agentic AI means **trusting coordination** and **accepting shared responsibility between humans and machines**

As organizations move up the stack, they need to change how they **govern, monitor, and own** these systems

Artificial Intelligence is the easy part. **Designing for trust is the real work.** It demands better decisions and guaranteed actions



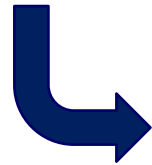
Creators, Providers and Users: *Different visibilities*



The human choices of **creators** and **data providers** have not disappeared

- 1) Selection of DATA sources
- 2) Selection of some types of DATA
- 3) Choice of MODELS and their parameters
- 4) Choice of RESULTS to be shown

The choices are masked from **users** by rhetoric about the *end of theory* and *greater precision* of algorithms compared to humans



IN MACHINES WE (must) TRUST

Any possible analysis and critics about the choices of data sources, models, shown results is prevented



Obscure (to us) choices

Considering the percentage of failures of the AI projects of the last 15 years, the principle “the numbers speak for themselves” appears more like an ambitious project for a new cognitive paradigm than a truly functioning reality

Nevertheless, today we are oriented (or forced) to accept:

- black-boxes instead of open models
- non-explainability of deep learning and of GenAI
- errors that are nicely defined *hallucinations*



Always ask yourself: when are they acceptable?



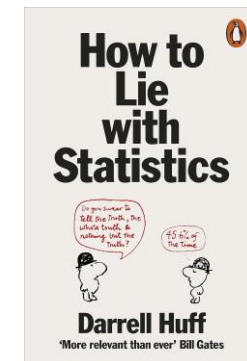
Key importance of results presentation



«We tell stories about the data and essentially they are the stories we wish to tell»

(Farida Vis, “A critical reflection on Big Data”, *First Monday*, 2013)

Era of “probabilistic truth” and of “alternate fact”



Some pessimistic opinions

- “Although AGI is far, we are heading into a situation in which systems will be making choices for us. And we should stop and ask ourselves what happens when those systems put aside human strategy in favor of something totally unknown to us”
- “If we don’t change this trajectory, we could be headed straight for catastrophe. But there is still time to act, and a role for everyone to play”

[The Big Nine: How the Tech Titans and Their Thinking Machines Could Warp Humanity, Amy Webb, futurist, NYU]



The main human challenges

AI is a versatile tool that can help people and organizations address many challenges, but it is not enough on its own

Without people that *identify*, *prioritize*, *design*, *evaluate* problems and *present solutions*, AI will have little impact and, at worst, can cause negative consequences

For each of these actions, humans remain the (secret) weapon for a brilliant future



The ethics of advanced AI assistants

Advanced AI assistants could have a profound impact on users and society, and be integrated into most aspects of people's lives. People may ask them to book holidays, manage social time or perform other life tasks. AI assistants could impact the way people approach work, education, creative projects, hobbies and social interaction

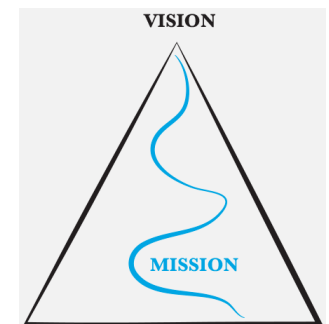
Over time, AI assistants could also influence the goals people pursue and their path of personal development through the information and advice assistants give and the actions they take.

With more autonomy, AI assistants present challenges and greater risk of:

- **accidents** caused by unclear or misinterpreted instructions
- **misalignment** with the user's values and interests
- **misuse**, like spreading misinformation or engaging in cyber attacks

Limits must be set on this technology.

Their values must better align to human values and be compatible with wider societal ideals and standards



First example: The ethics of advanced AI assistants

- Imagine a future where we interact regularly with a range of advanced AI assistants, and where millions of assistants interact with each other on our behalf. **These experiences and interactions may soon become part of our everyday reality**
- AI assistants will be capable of **planning and performing a wide range of actions in line with a person's aims**. They could add immense value to people's lives and to society, serving as **creative partners, research analysts, educational tutors, life planners and more**
- This is why it is so important to think proactively about what this world could look like, and to help steer responsible decision-making and beneficial outcomes ahead of time

<https://storage.googleapis.com/deepmind-media/DeepMind.com/Blog/ethics-of-advanced-ai-assistants/the-ethics-of-advanced-ai-assistants-2024-i.pdf>

(19 April 2024 - 274 pages)



<https://www.youtube.com/watch?v=dNj-kbyxem0> (93 minutes)



Second example: NIST - AI Risk Management Framework

(<https://doi.org/10.6028/NIST.AI.100-1>)

- **Social responsibility** can refer to the organization's responsibility "for the impacts of its decisions and activities on society and the environment through transparent and ethical behavior" (ISO 26000:2010).
- **Sustainability** refers to the "state of the global system, including environmental, social, and economic aspects, in which the needs of the present are met without compromising the ability of future generations to meet their own needs" (ISO/IEC TR 24368:2022)
- **Responsible AI** is meant to result in technology that is also equitable and accountable. The expectation is that organizational practices are carried out in accord with "professional responsibility," defined by ISO as an approach that "aims to ensure that professionals who design, develop, or deploy AI systems and applications or AI-based products or systems, recognize their unique position to exert influence on people, society, and the future of AI" (ISO/IEC TR 24368:2022)



“What do you mean we don’t know exactly how *deep learning* works if it was designed by people?”

“No. It wasn’t.”

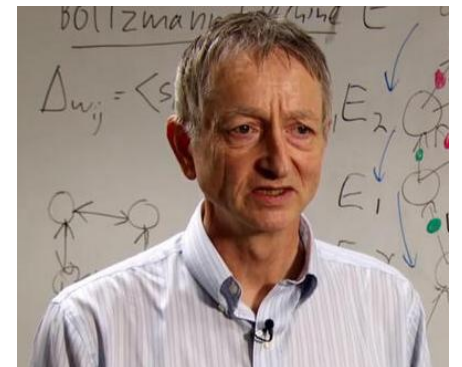
“We have a rough idea of what it’s doing, but as soon as training gets really complicated, **we don’t actually know what’s going on any more than we know what is going on in your brain”**

We designed the learning algorithm that is like designing the principle of evolution. But when this learning algorithm interacts with data, it produces complicated neural networks that are good at doing things, **but we don’t really understand exactly how they do these things”**

“Right now, they are not smarter than us, as far as I know. But I think they could be”



Geoffrey Hinton,
British computer
scientist, defined
"the Godfather of AI"

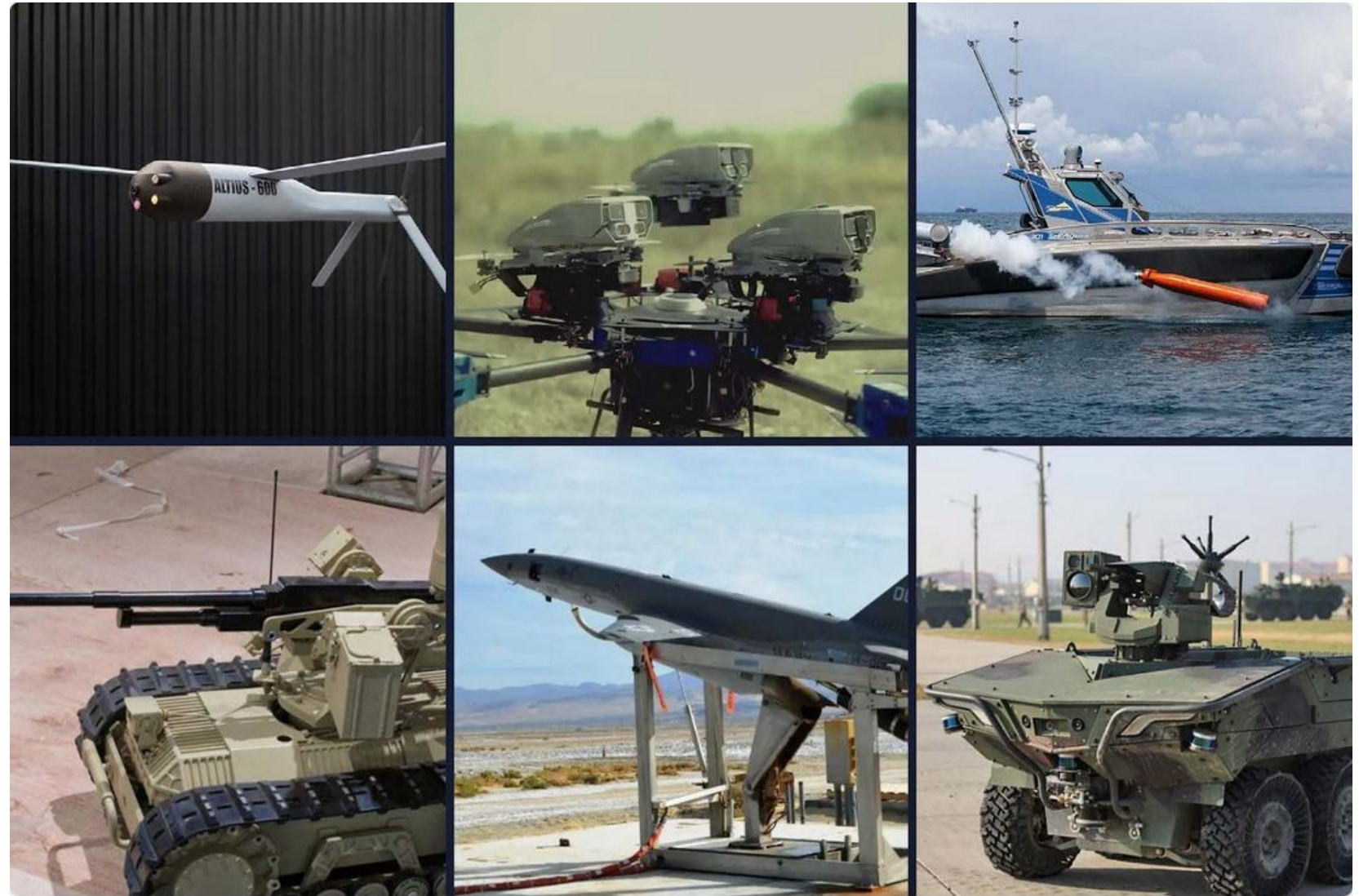
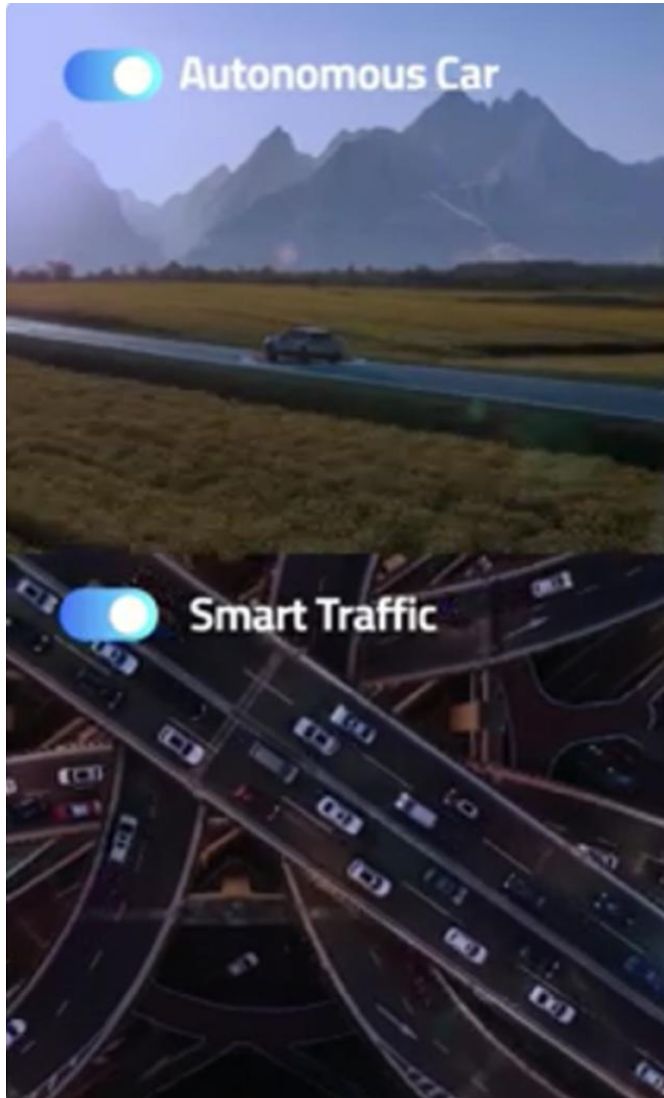


Can we trust something that we don’t know how it works?

Nevertheless, we are giving it autonomy (*in little things*)



Nevertheless, we are giving it autonomy (*in serious things*)



Autonomous weapons that are available on the market today

Source: autonomousweaponswatch.org



An unexpected(?) strategic change of some providers

Recent declarations

«In many sectors (e.g., healthcare), no AI service will be created and managed without the support of humans with the appropriate skills»

«It is essential to keep a human being at the center of everything that happens with AI and technology»

«Workers cannot be replaced by AI, but thanks to AI they will be able to be better informed and more efficient»



Unclear motivations

Technological failures?

Marketing?

Legal liabilities?

Norms?

Insurance issues?

Others?

Many examples leveraging GenAI

The Intelligent Copilot

How Generative AI will transform the future smart cockpit



From replacement risks to collaboration



Don't forget that, while it helps you, it learns from you (and other million of people). *Future scenario?*

Human role

Human skills remain indispensable

Jobs are not at real risk.

Some **tasks** will change; many tasks will be enhanced.

(In countries subject to decline in birth rates, AI even might represent a key solution)

There are and will be **no technological magic pills** that will solve all the problems for us



Cognitive debt [MIT]

- MIT measures what we lose using AI only: just 16% of students remember what they wrote with AI
 - ➔ **Cognitive debt builds up silently, like a muscle that stops working**



Double check

- In our study, we tested something concrete: **A model can obey the rules when everything is normal, but if someone manipulates the conversation in the right way, the model can “follow the thread of the conversation” instead of stopping to double-check permissions.** To avoid disrupting the coherence of the response, the model risks complying with a request it should not. Not because it has been “hacked,” but because it is designed to complete texts fluidly and coherently
- In some cases, narrative coherence can outweigh the rules. A language model can assist in processes, but **it cannot be the ultimate guarantor of security. The real checks must reside outside the model, in a separate, non-circumventable layer.** If we bring AI into critical systems, we must keep in mind that **it was not designed to be a gatekeeper**



(Important document: “Strategia 2026 sull’Intelligenza Artificiale”)

AI is not a “tech project,” it is a governance capability that impacts sovereignty, freedom of action, and credibility among the partners (e.g., within the NATO/EU context)

Three key messages:

- 1) **The value of data: quality + trust + lifecycle**: “Data is the fundamental strategic asset. It is valuable if it is reliable, governed, protected, and has verifiable provenance and traceability”
- 2) **Decision-making superiority**. In an increasingly transparent and multi-domain battlefield, the **winner is the one who transforms data into decisions first**, reducing the time between detection, assessment (C2), and operational response
- 3) **Strategic autonomy** (*decision-making sovereignty*). Superiority is not about having advanced tools, but about **transforming information and analysis into action** while maintaining continuity, control, and accountability of command. This also requires reducing critical external dependencies: AI must remain governable and available when truly needed, not just when everything is functioning “in ideal conditions.”

